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(54) **REGISTRATION AND PARALLAX ERROR CORRECTION FOR VIDEO SEE-THROUGH (VST) EXTENDED REALITY (XR)**

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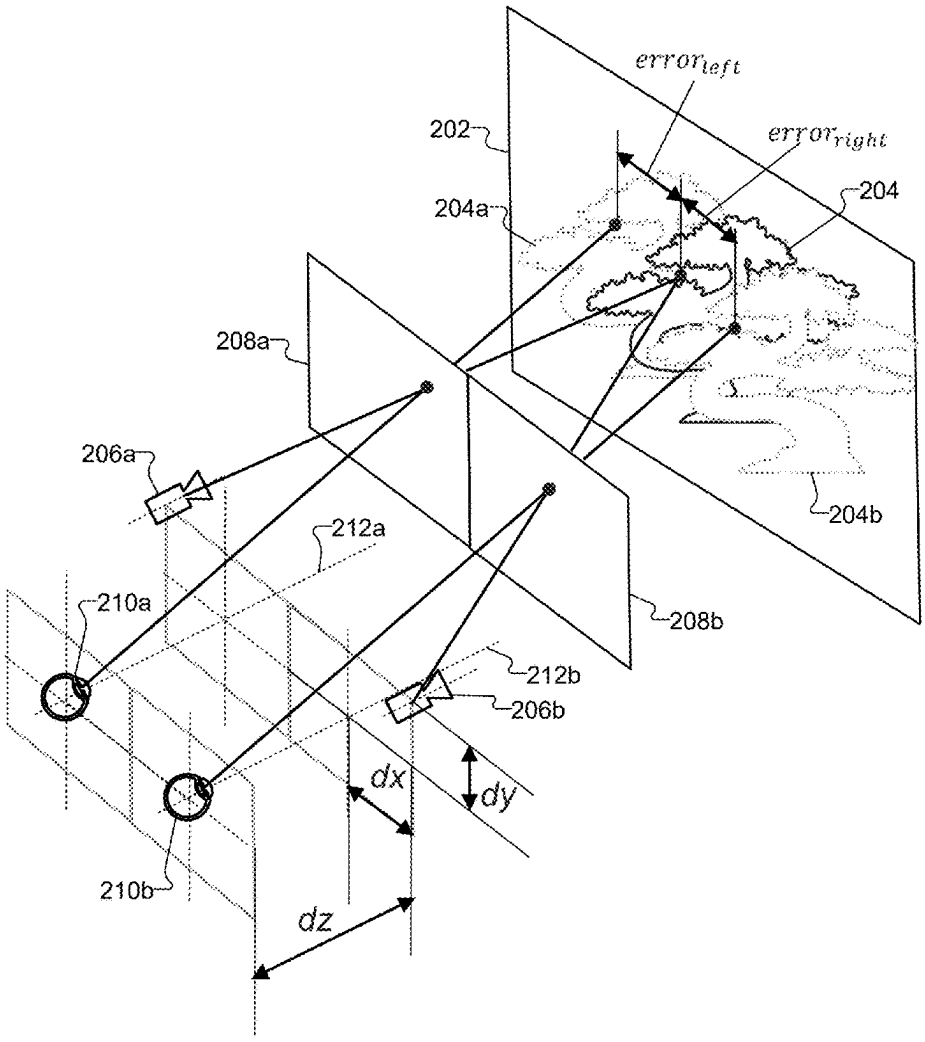
CPC ***G06T 11/60*** (2013.01); ***G06T 7/80*** (2017.01)

(57) **ABSTRACT**

A method includes identifying a transformation associated with a video see-through (VST) extended reality (XR) device using registration and parallax errors. The registration and parallax errors are based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel. The method also includes obtaining an image that captures an object using the see-through camera, applying the transformation to the image in order to generate a modified image, and rendering the modified image for presentation on the display panel. Applying the transformation modifies the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.



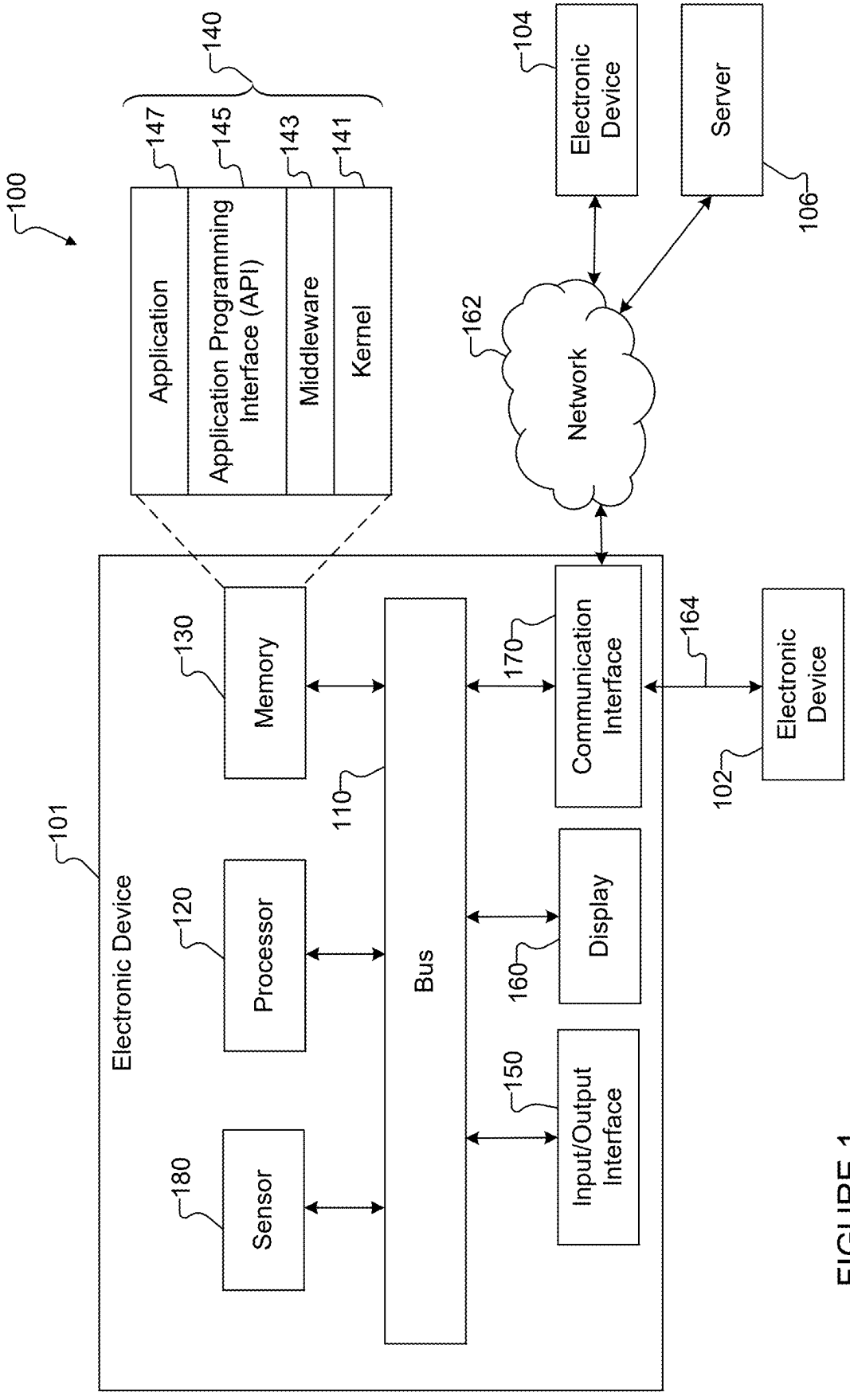

100
140
147
145
143
141
Application
Application Programming Interface (API)
Middleware
Kernel
104
Electronic Device
Server
106
162
Network
101
Electronic Device
130
Memory
110
Bus
170
Communication Interface
164
Electronic Device
102
120
Processor
160
Display
150
Input/Output Interface
180
Sensor

FIGURE 1

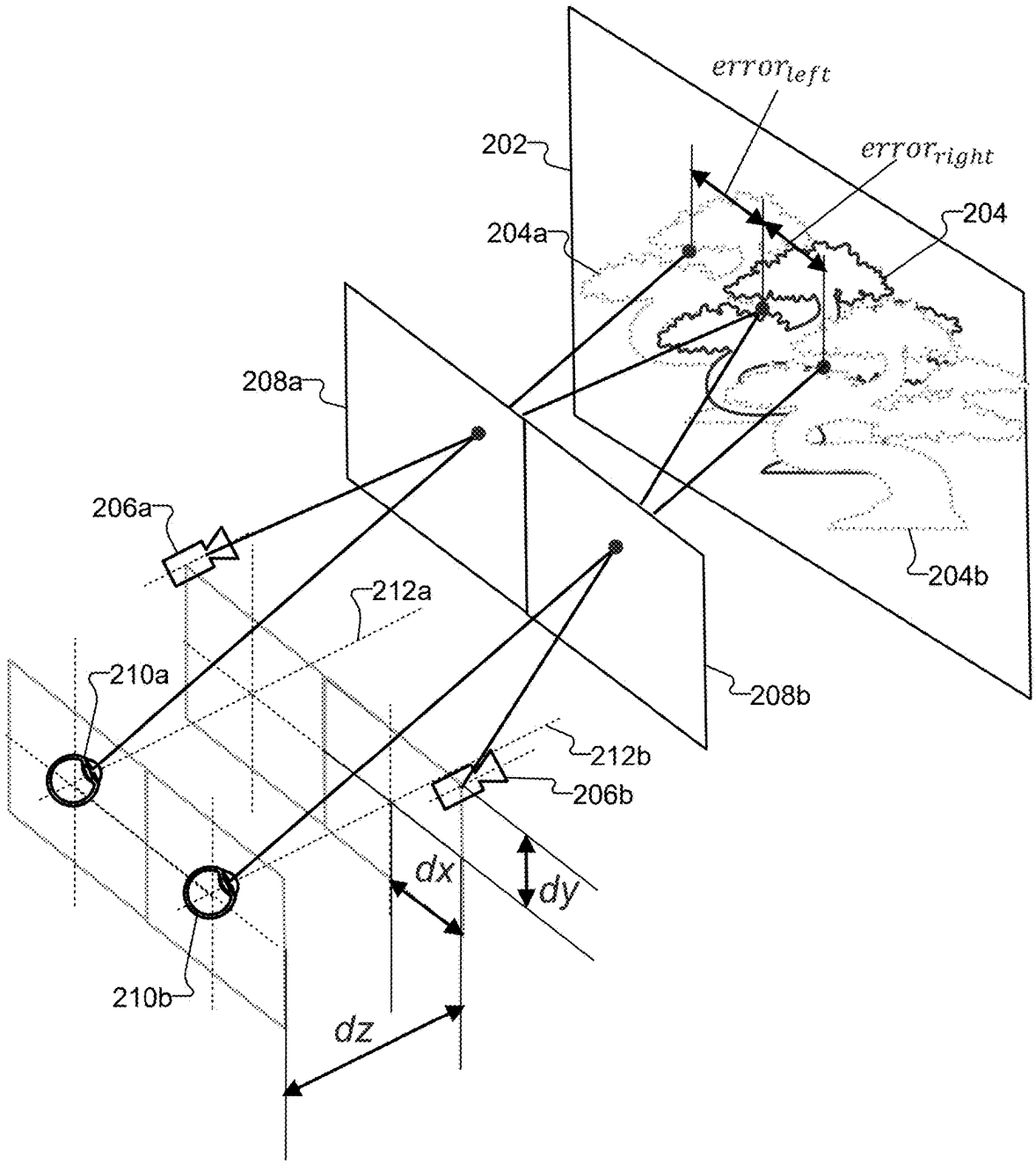

$error_{left}$
$error_{right}$
202
204
204a
208a
206a
212a
210a
204b
208b
212b
206b
dx
dy
210b
dz

FIGURE 2A

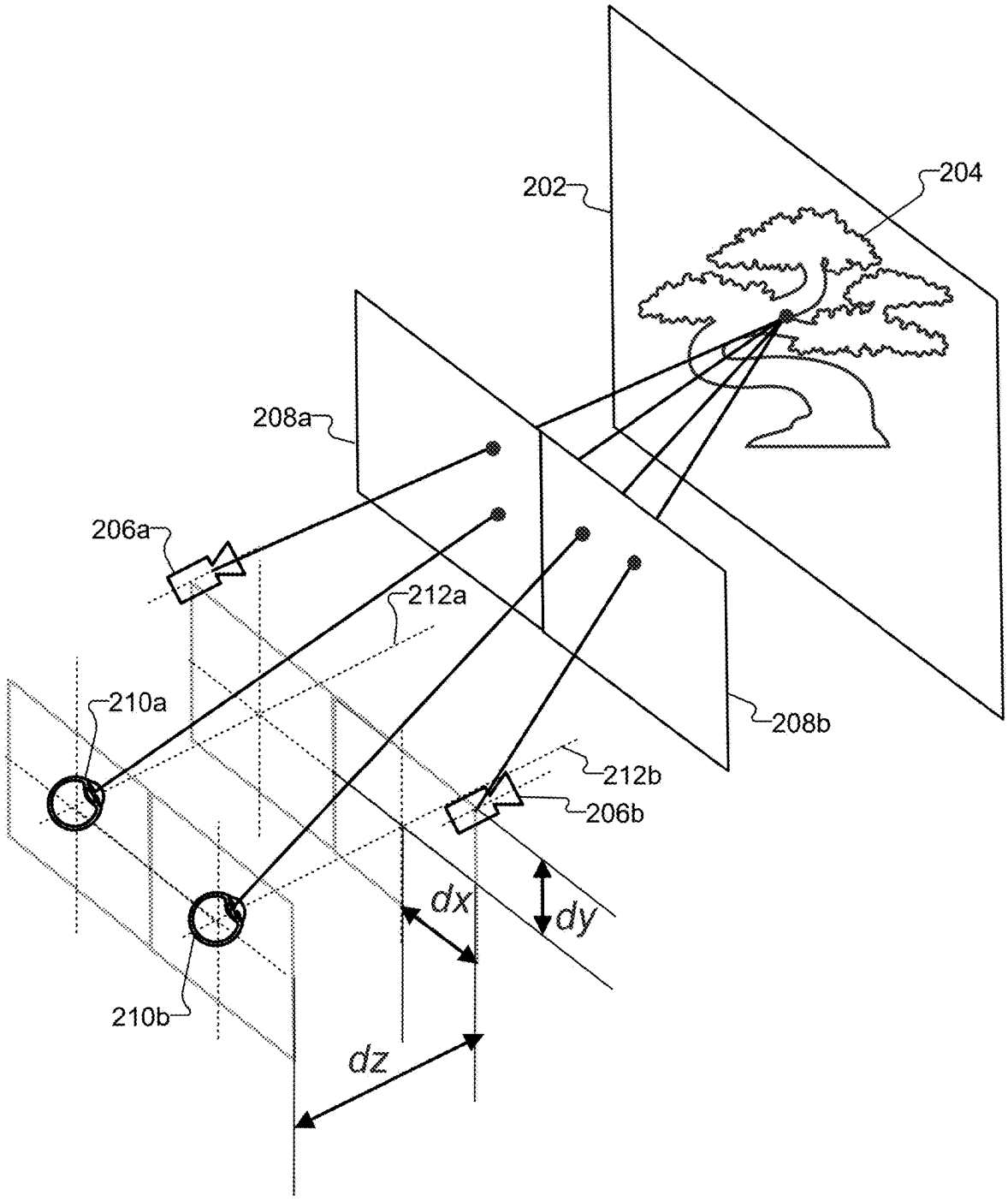


FIGURE 2B

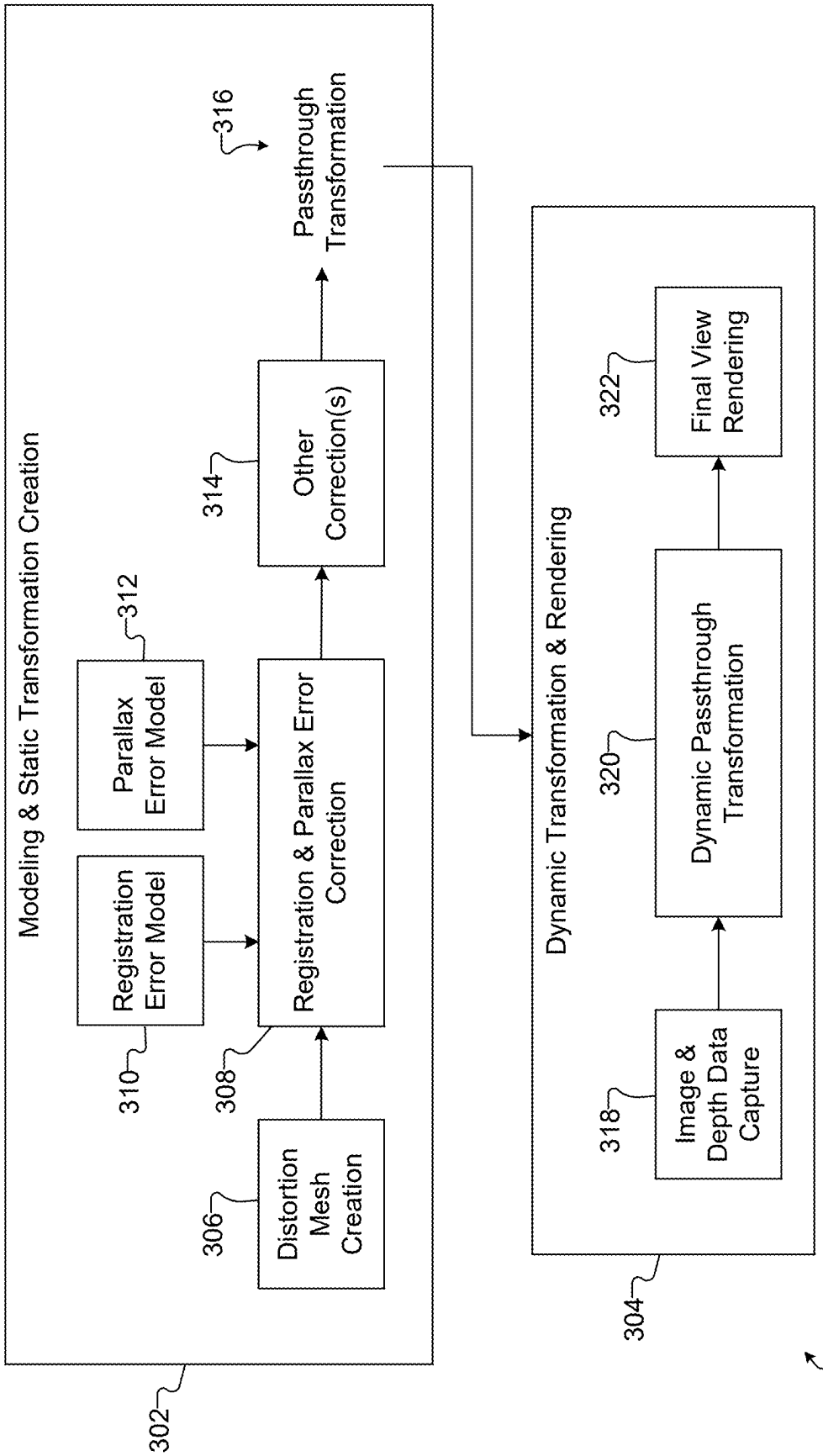

Modeling & Static Transformation Creation
Distortion Mesh Creation
306
Registration Error Model
310
Parallax Error Model
312
Registration & Parallax Error Correction
308
Other Correction(s)
314
Passthrough Transformation
316
302
Dynamic Transformation & Rendering
Image & Depth Data Capture
318
Dynamic Passthrough Transformation
320
Final View Rendering
322
304
300

FIGURE 3

FIGURE 4A

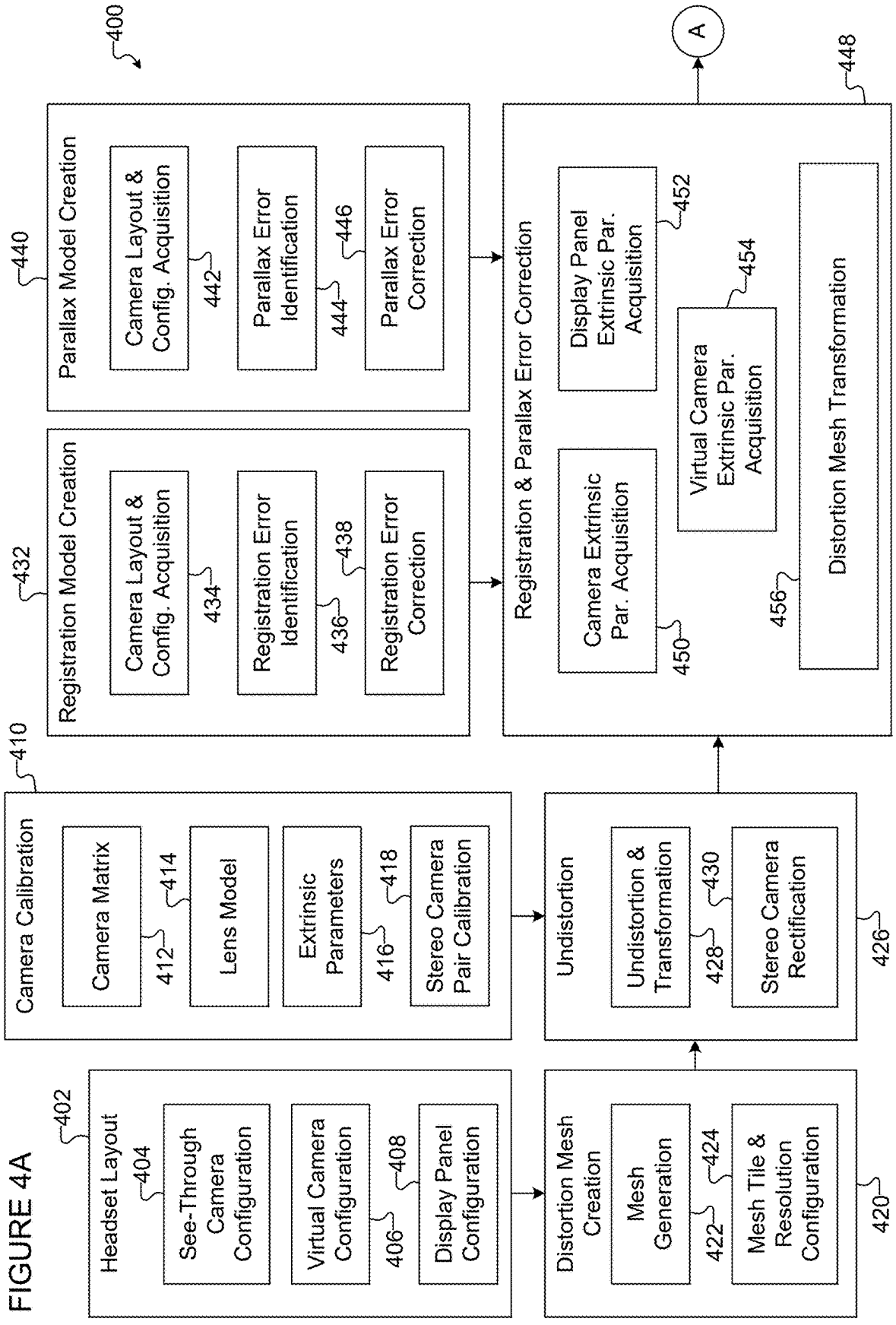
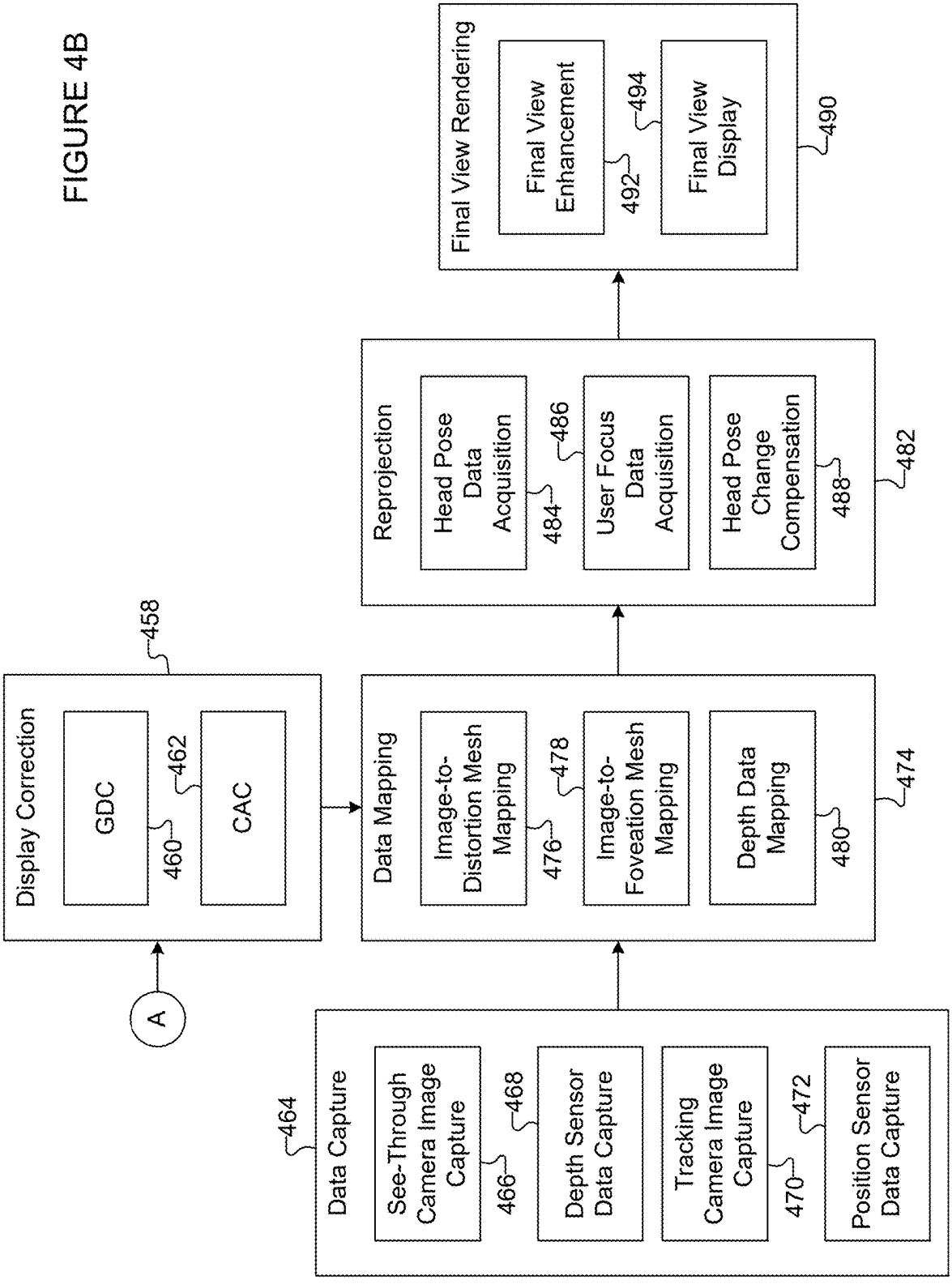

400
402
Headset Layout
404
See-Through Camera Configuration
Virtual Camera Configuration
406
408
Display Panel Configuration
410
Camera Calibration
Camera Matrix
412
414
Lens Model
Extrinsic Parameters
416
418
Stereo Camera Pair Calibration
420
Distortion Mesh Creation
Mesh Generation
422
424
Mesh Tile & Resolution Configuration
426
Undistortion
Undistortion & Transformation
428
430
Stereo Camera Rectification
432
Registration Model Creation
Camera Layout & Config. Acquisition
434
Registration Error Identification
436
438
Registration Error Correction
440
Parallax Model Creation
Camera Layout & Config. Acquisition
442
Parallax Error Identification
444
446
Parallax Error Correction
448
Registration & Parallax Error Correction
Camera Extrinsic Par. Acquisition
450
Display Panel Extrinsic Par. Acquisition
452
Virtual Camera Extrinsic Par. Acquisition
454
456
Distortion Mesh Transformation
A

FIGURE 4B


Final View Rendering
Final View Enhancement
492
494
Final View Display
490
Reprojection
Head Pose Data Acquisition
484
486
User Focus Data Acquisition
Head Pose Change Compensation
488
482
Display Correction
458
GDC
460
462
CAC
Data Mapping
Image-to-Distortion Mesh Mapping
476
478
Image-to-Foveation Mesh Mapping
Depth Data Mapping
480
474
A
Data Capture
464
See-Through Camera Image Capture
466
468
Depth Sensor Data Capture
Tracking Camera Image Capture
470
472
Position Sensor Data Capture

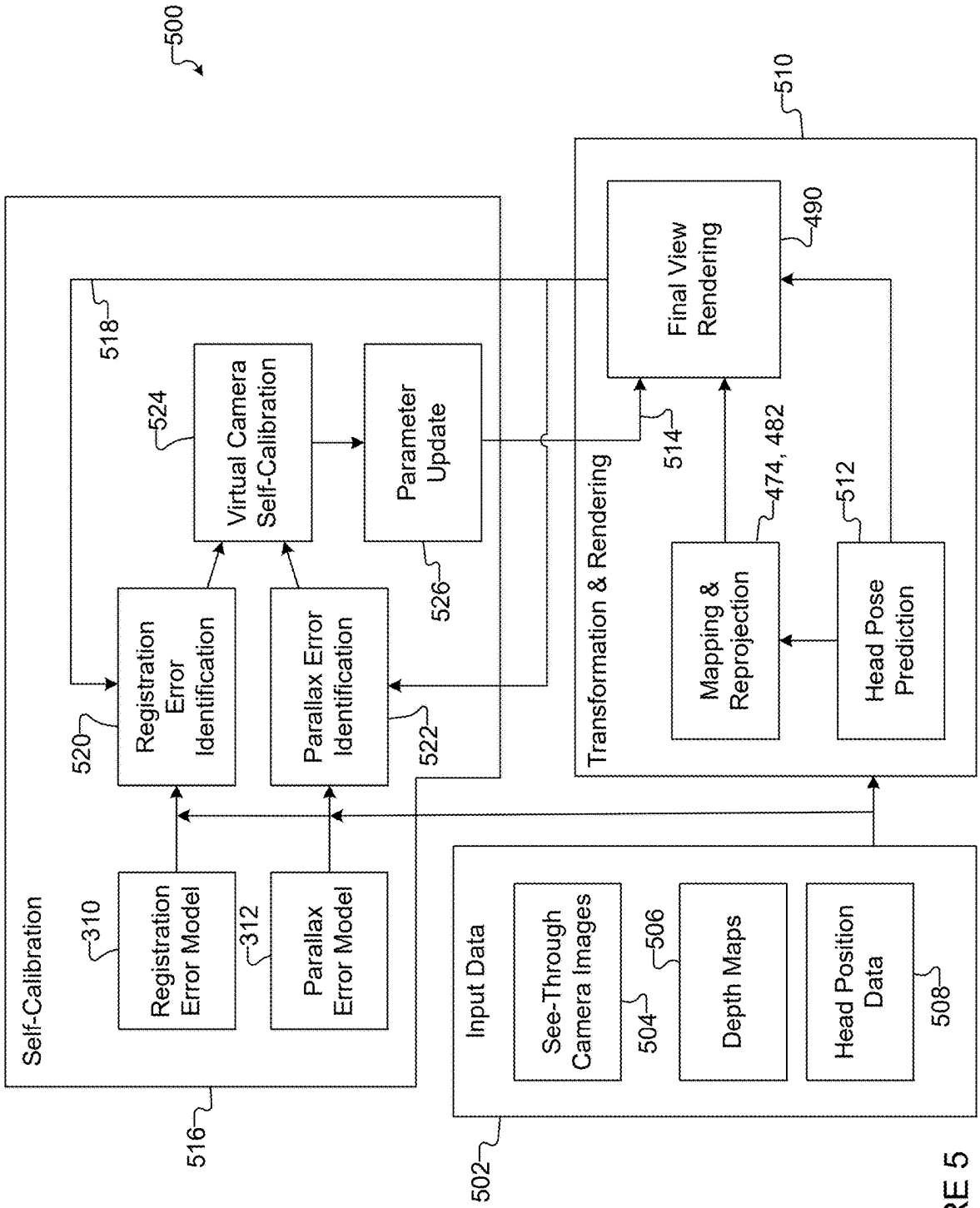

500
Self-Calibration
310
Registration Error Model
312
Parallax Error Model
516
518
520
Registration Error Identification
522
Parallax Error Identification
524
Virtual Camera Self-Calibration
526
Parameter Update
Input Data
See-Through Camera Images
504
506
Depth Maps
Head Position Data
508
502
Transformation & Rendering
510
Final View Rendering
490
514
474, 482
Mapping & Reprojection
512
Head Pose Prediction

FIGURE 5

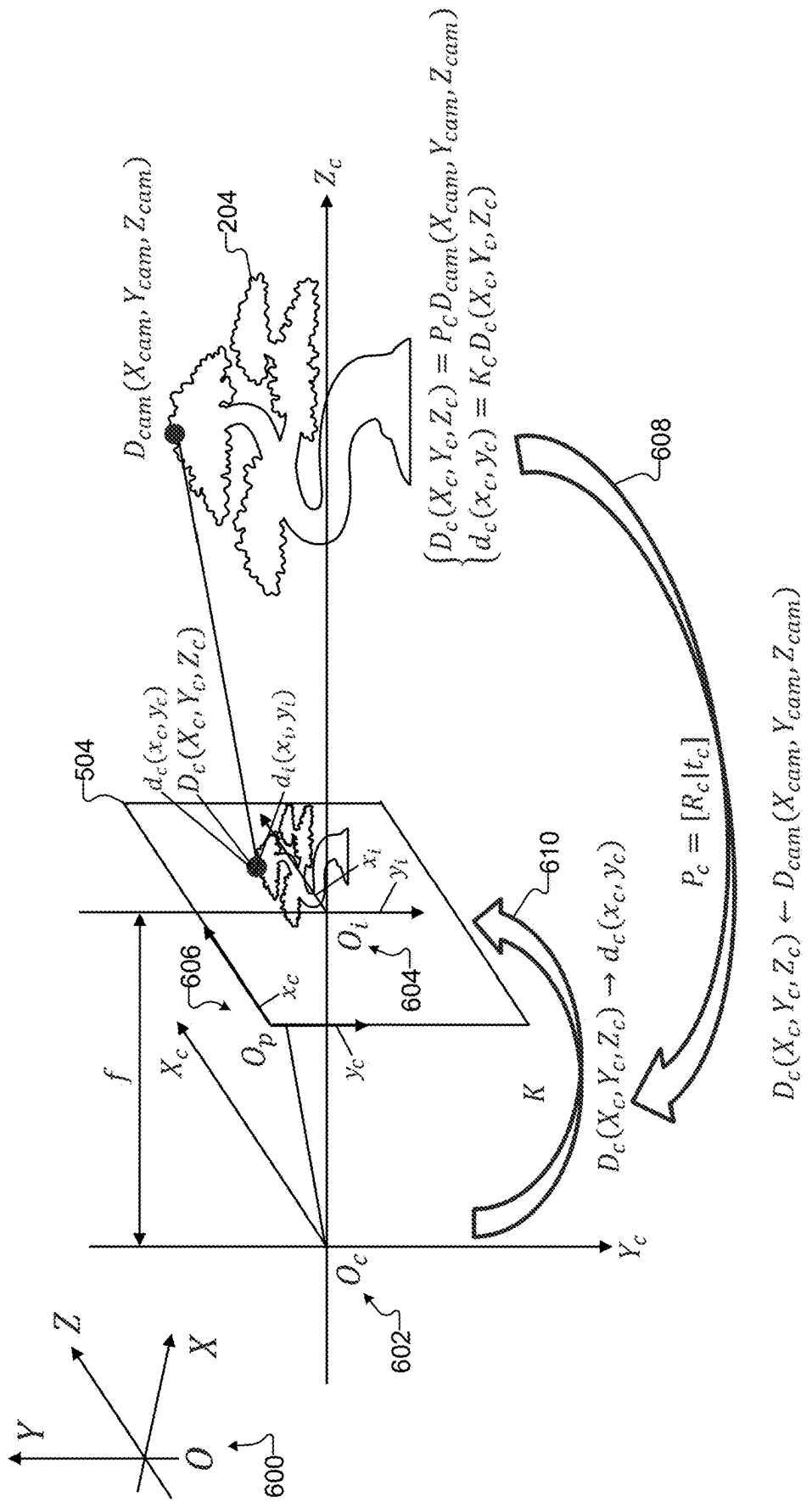

$D_{cam}(X_{cam}, Y_{cam}, Z_{cam})$
204
Z_c
$\begin{cases} D_c(X_c, Y_c, Z_c) = P_c D_{cam}(X_{cam}, Y_{cam}, Z_{cam}) \\ d_c(x_c, y_c) = K_c D_c(X_c, Y_c, Z_c) \end{cases}$
608
504
$d_c(x_c, y_c)$
$D_c(X_c, Y_c, Z_c)$
$d_i(x_i, y_i)$
x_i
y_i
610
$P_c = [R_c|t_c]$
O_i
604
606
x_c
O_p
X_c
f
y_c
K
$D_c(X_c, Y_c, Z_c) \rightarrow d_c(x_c, y_c)$
Y_c
O_c
602
$D_c(X_c, Y_c, Z_c) \leftarrow D_{cam}(X_{cam}, Y_{cam}, Z_{cam})$
Z
X
Y
O
600

FIGURE 6

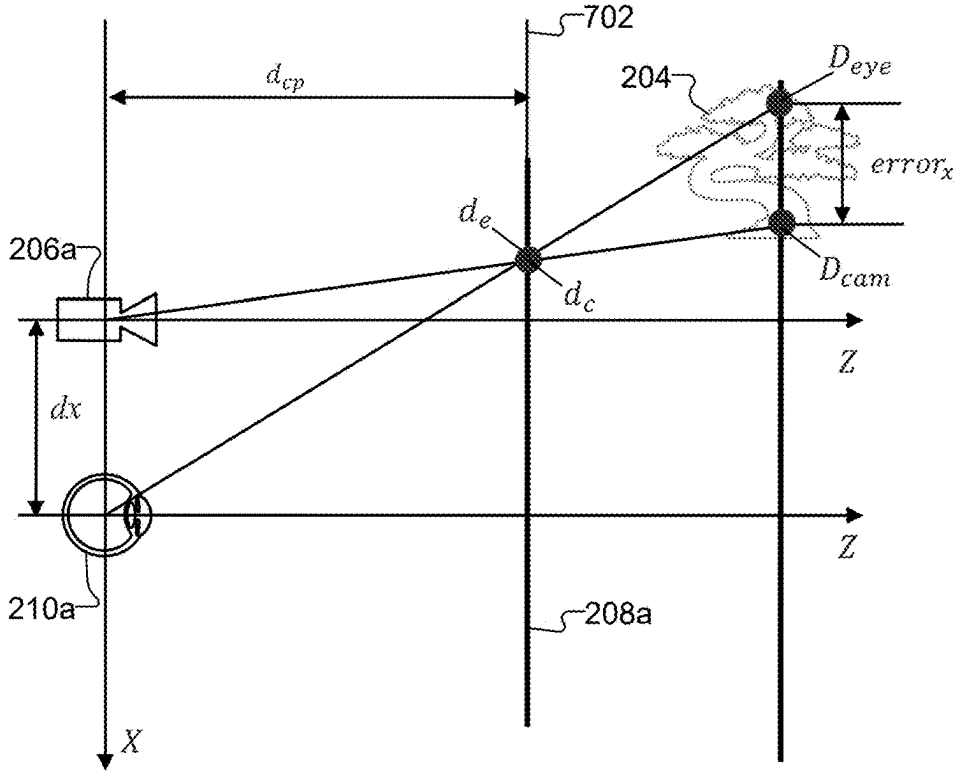


FIGURE 7

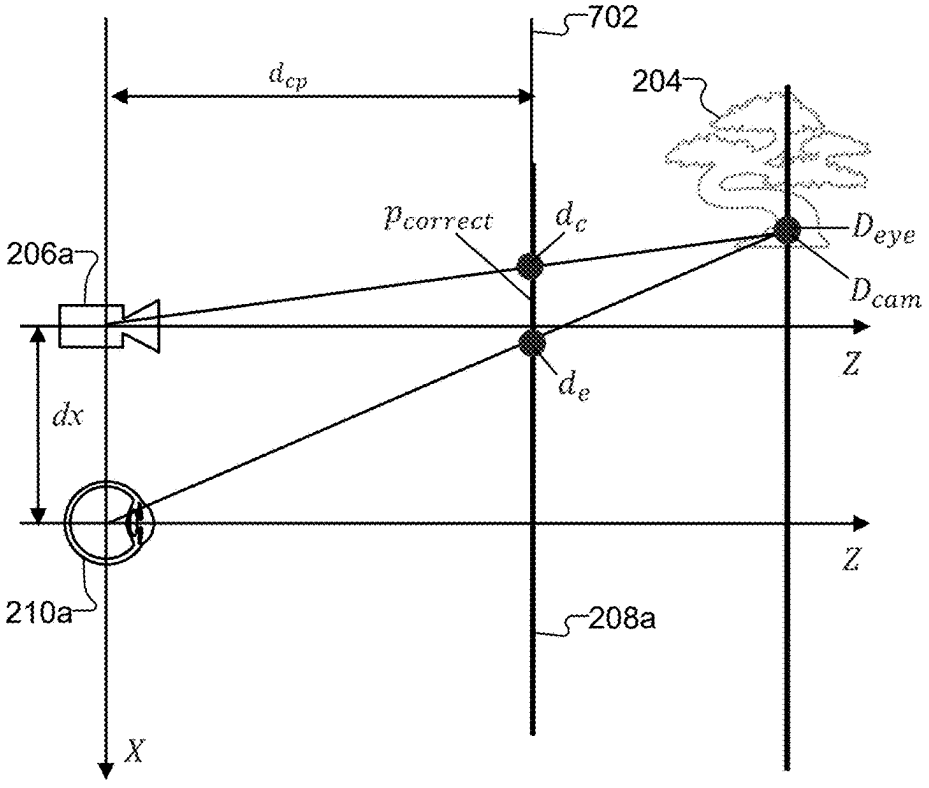


FIGURE 8

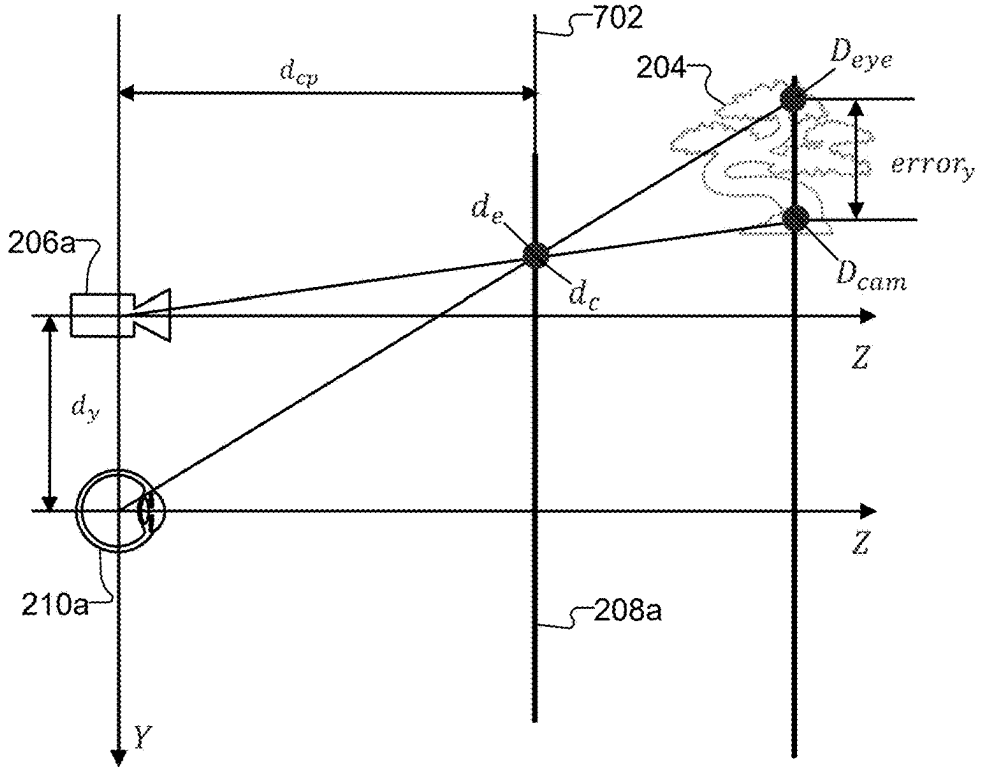


FIGURE 9

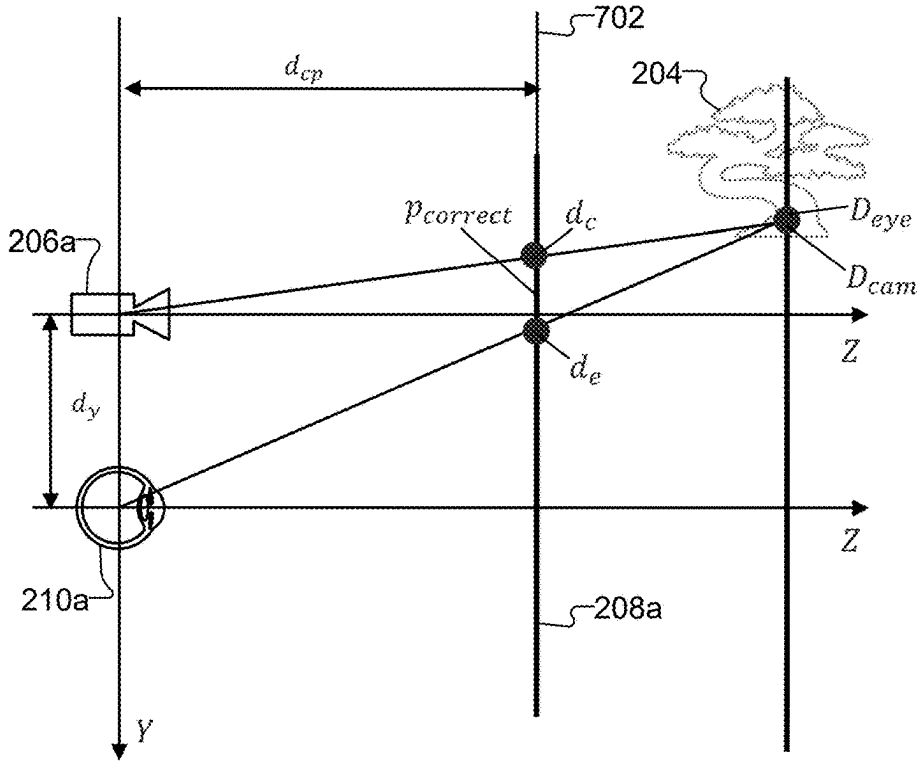


FIGURE 10

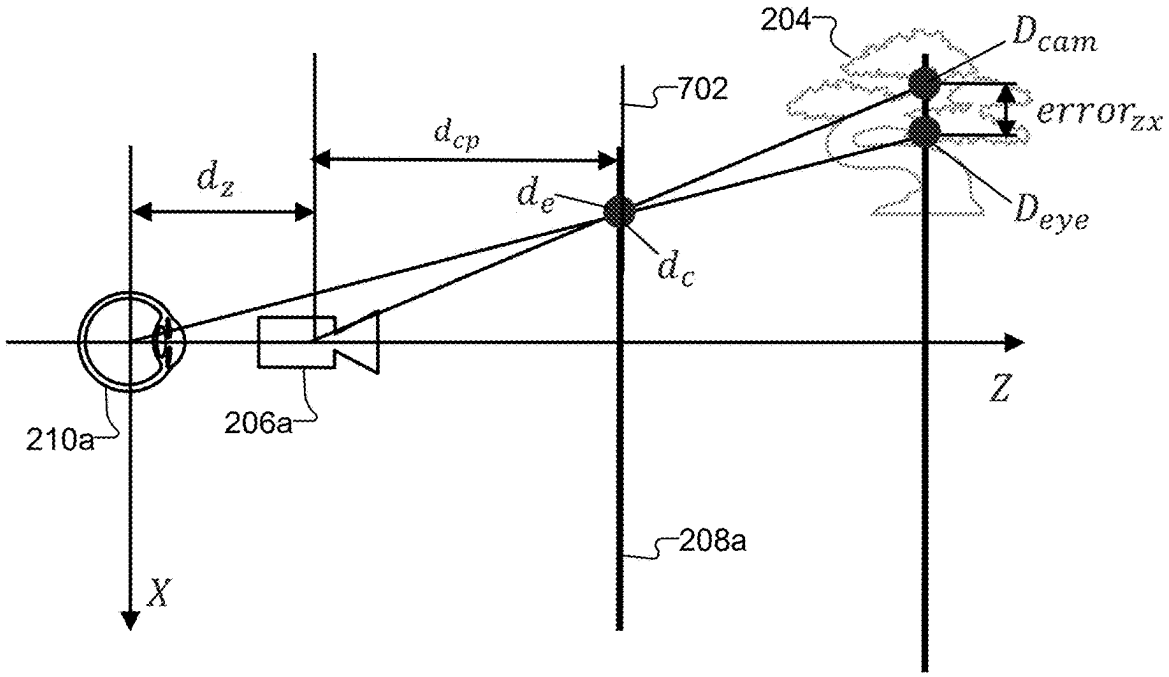


FIGURE 11

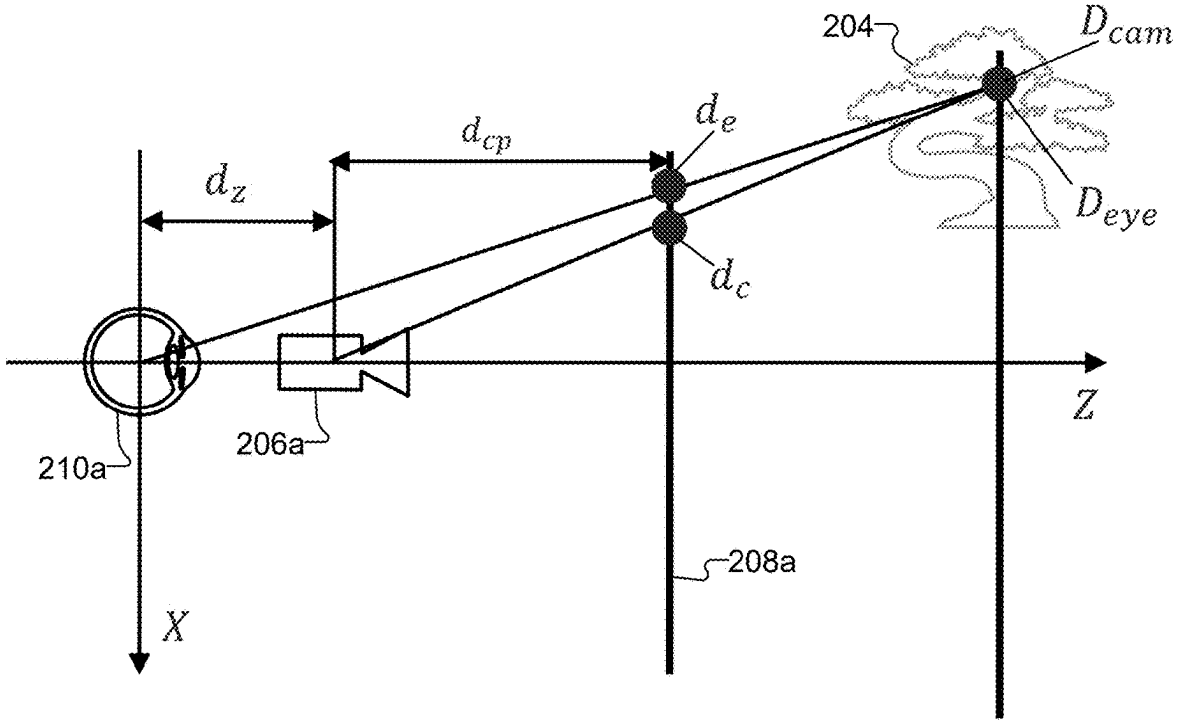


FIGURE 12

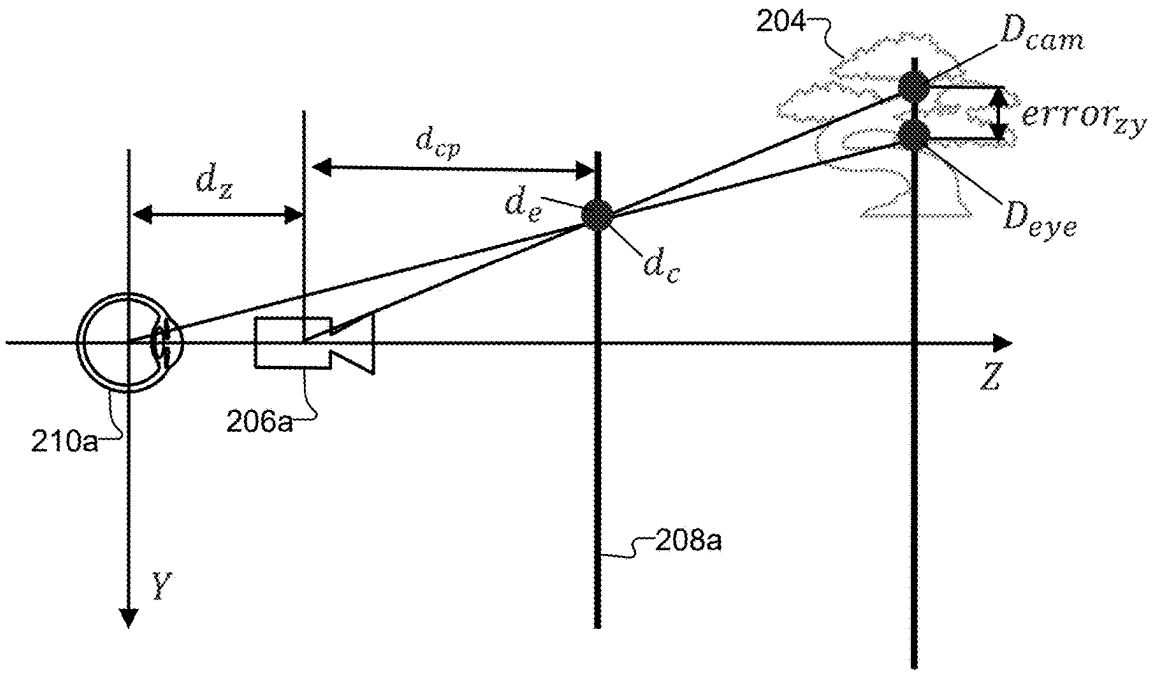


FIGURE 13

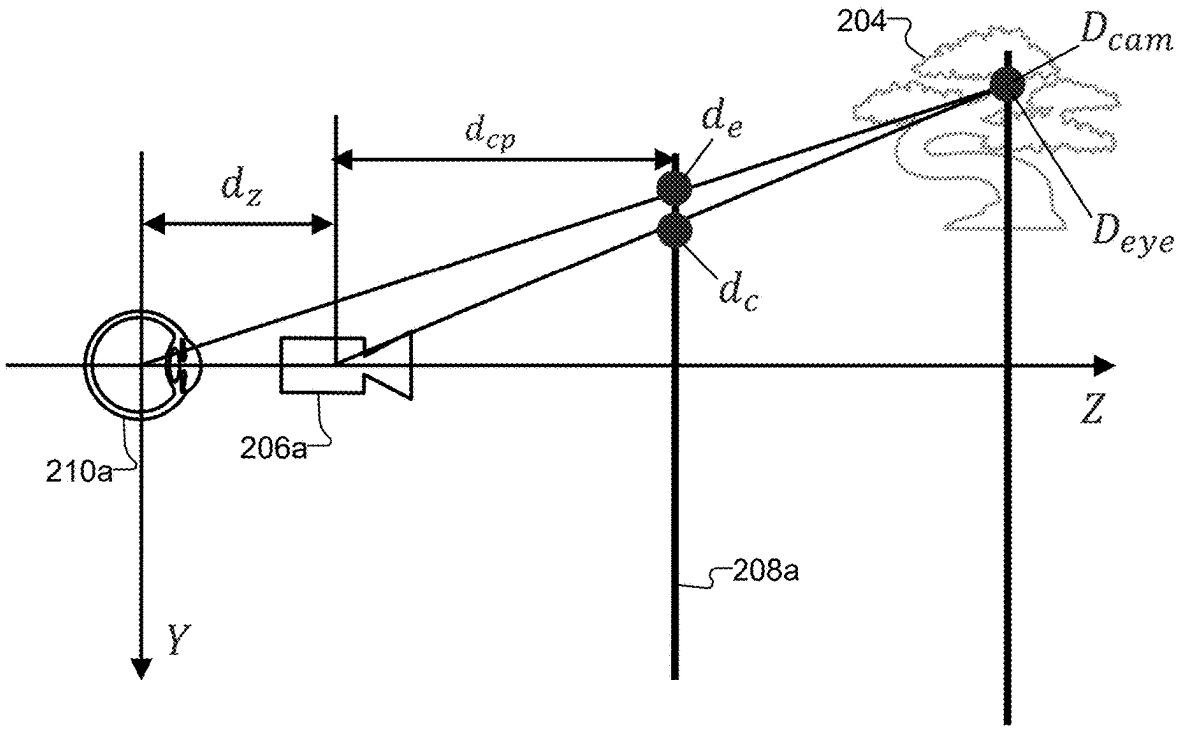


FIGURE 14

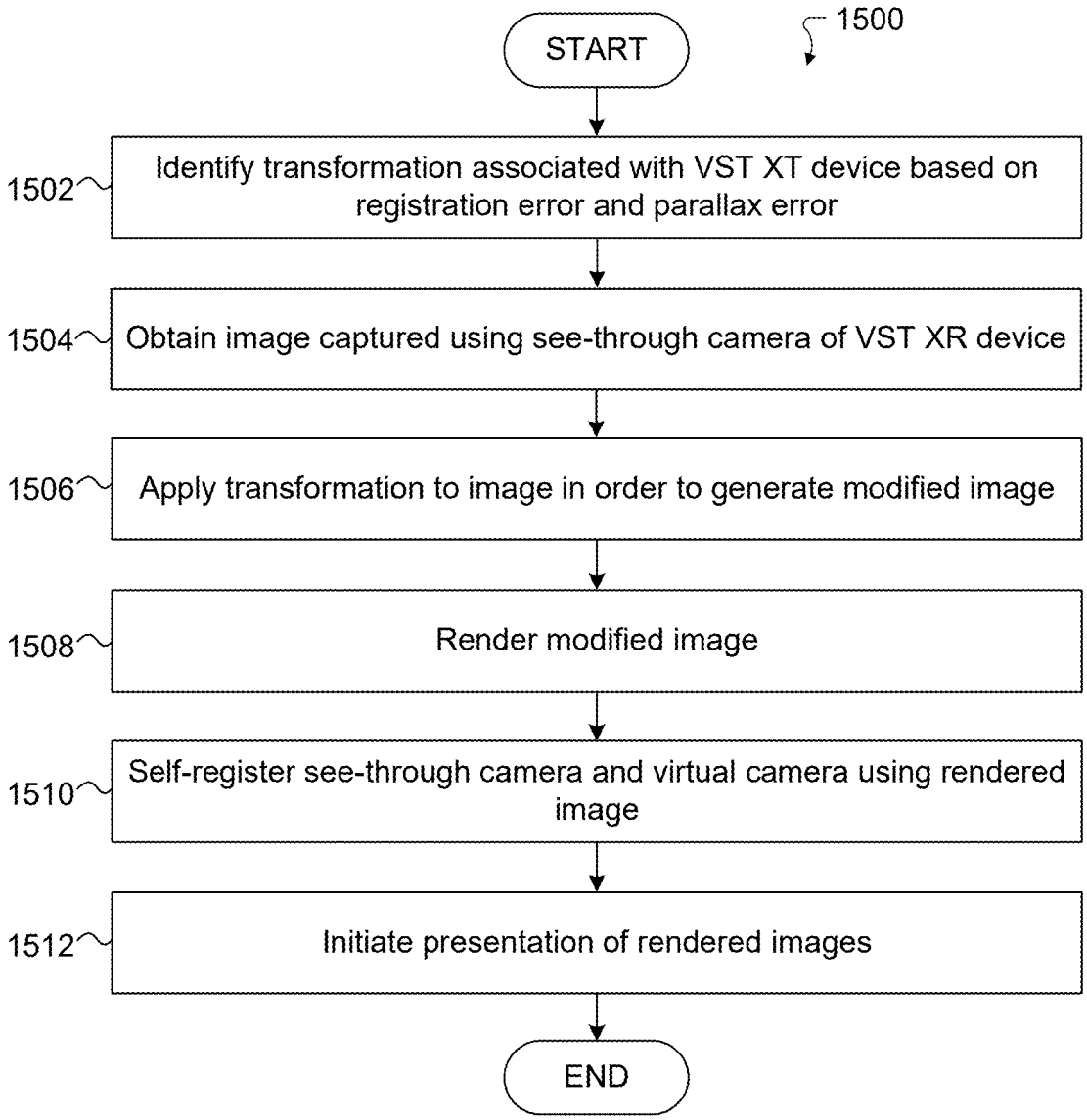

1500
START
1502
Identify transformation associated with VST XT device based on registration error and parallax error
1504
Obtain image captured using see-through camera of VST XR device
1506
Apply transformation to image in order to generate modified image
1508
Render modified image
1510
Self-register see-through camera and virtual camera using rendered image
1512
Initiate presentation of rendered images
END

FIGURE 15

REGISTRATION AND PARALLAX ERROR CORRECTION FOR VIDEO SEE-THROUGH (VST) EXTENDED REALITY (XR)

CROSS-REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM

[0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/559,143 filed on Feb. 28, 2024. This provisional patent application is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to extended reality (XR) systems and processes. More specifically, this disclosure relates to registration and parallax error correction for video see-through (VST) XR.

BACKGROUND

[0003] Extended reality (XR) systems are becoming more and more popular over time, and numerous applications have been and are being developed for XR systems. Some XR systems (such as augmented reality or "AR" systems and mixed reality or "MR" systems) can enhance a user's view of his or her current environment by overlaying digital content (such as information or virtual objects) over the user's view of the current environment. For example, some XR systems can often seamlessly blend virtual objects generated by computer graphics with real-world scenes.

SUMMARY

[0004] This disclosure relates to registration and parallax error correction for video see-through (VST) extended reality (XR).

[0005] In a first embodiment, a method includes identifying a transformation associated with a VST XR device using a registration error and a parallax error. The registration error and the parallax error are based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera of the VST XR device and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel of the VST device. The method also includes obtaining an image that captures an object using the see-through camera, applying the transformation to the image in order to generate a modified image, and rendering the modified image for presentation on the display panel. Applying the transformation modifies the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

[0006] In a second embodiment, a VST XR device includes a see-through camera, a display panel, and at least one processing device. The at least one processing device is configured to identify a transformation using a registration error and a parallax error. The registration error and the parallax error are based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using the see-through camera and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing the display panel. The at least one processing device is also configured to obtain an image that captures an object using the see-through camera, apply the transformation to the image in order to generate a modified image, and render the modified image for presentation on the display panel. The at least one processing device is configured to apply the transformation in order to modify the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

[0007] In a third embodiment, a non-transitory machine readable medium contains instructions that when executed cause at least one processor of a VST XR device to identify a transformation using a registration error and a parallax error. The registration error and the parallax error are based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera of the VST XR device and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel of the VST device. The non-transitory machine readable medium also contains instructions that when executed cause the at least one processor to obtain an image that captures an object using the see-through camera, apply the transformation to the image in order to generate a modified image, and render the modified image for presentation on the display panel. The instructions when executed cause the at least one processor to apply the transformation in order to modify the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

[0008] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

[0009] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms "transmit," "receive," and "communicate," as well as derivatives thereof, encompass both direct and indirect communication. The terms "include" and "comprise," as well as derivatives thereof, mean inclusion without limitation. The term "or" is inclusive, meaning and/or. The phrase "associated with," as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like.

[0010] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms "application" and "program" refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase "computer readable program code" includes any type of computer code, including source code, object code, and executable code. The phrase "computer readable medium" includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A "non-transitory" computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other

signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0011] As used here, terms and phrases such as "have," "may have," "include," or "may include" a feature (like a number, function, operation, or component such as a part) indicate the existence of the feature and do not exclude the existence of other features. Also, as used here, the phrases "A or B," "at least one of A and/or B," or "one or more of A and/or B" may include all possible combinations of A and B. For example, "A or B," "at least one of A and B," and "at least one of A or B" may indicate all of (1) including at least one A, (2) including at least one B, or (3) including at least one A and at least one B. Further, as used here, the terms "first" and "second" may modify various components regardless of importance and do not limit the components. These terms are only used to distinguish one component from another. For example, a first user device and a second user device may indicate different user devices from each other, regardless of the order or importance of the devices. A first component may be denoted a second component and vice versa without departing from the scope of this disclosure.

[0012] It will be understood that, when an element (such as a first element) is referred to as being (operatively or communicatively) "coupled with/to" or "connected with/to" another element (such as a second element), it can be coupled or connected with/to the other element directly or via a third element. In contrast, it will be understood that, when an element (such as a first element) is referred to as being "directly coupled with/to" or "directly connected with/to" another element (such as a second element), no other element (such as a third element) intervenes between the element and the other element.

[0013] As used here, the phrase "configured (or set) to" may be interchangeably used with the phrases "suitable for," "having the capacity to," "designed to," "adapted to," "made to," or "capable of" depending on the circumstances. The phrase "configured (or set) to" does not essentially mean "specifically designed in hardware to." Rather, the phrase "configured to" may mean that a device can perform an operation together with another device or parts. For example, the phrase "processor configured (or set) to perform A, B, and C" may mean a generic-purpose processor (such as a CPU or application processor) that may perform the operations by executing one or more software programs stored in a memory device or a dedicated processor (such as an embedded processor) for performing the operations.

[0014] The terms and phrases as used here are provided merely to describe some embodiments of this disclosure but not to limit the scope of other embodiments of this disclosure. It is to be understood that the singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise. All terms and phrases, including technical and scientific terms and phrases, used here have the same meanings as commonly understood by one of ordinary skill in the art to which the embodiments of this disclosure belong. It will be further understood that terms and phrases, such as those defined in commonly-used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined here. In some cases, the terms and phrases defined here may be interpreted to exclude embodiments of this disclosure.

[0015] Examples of an "electronic device" according to embodiments of this disclosure may include at least one of a smartphone, a tablet personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop computer, a netbook computer, a workstation, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, or a wearable device (such as smart glasses, a head-mounted device (HMD), electronic clothes, an electronic bracelet, an electronic necklace, an electronic accessory, an electronic tattoo, a smart mirror, or a smart watch). Other examples of an electronic device include a smart home appliance. Examples of the smart home appliance may include at least one of a television, a digital video disc (DVD) player, an audio player, a refrigerator, an air conditioner, a cleaner, an oven, a microwave oven, a washer, a dryer, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV box (such as SAMSUNG HOMESYNC, APPLETV, or GOOGLE TV), a smart speaker or speaker with an integrated digital assistant (such as SAMSUNG GALAXY HOME, APPLE HOMEPOD, or AMAZON ECHO), a gaming console (such as an XBOX, PLAYSTATION, or NINTENDO), an electronic dictionary, an electronic key, a camcorder, or an electronic picture frame. Still other examples of an electronic device include at least one of various medical devices (such as diverse portable medical measuring devices (like a blood sugar measuring device, a heartbeat measuring device, or a body temperature measuring device), a magnetic resource angiography (MRA) device, a magnetic resource imaging (MRI) device, a computed tomography (CT) device, an imaging device, or an ultrasonic device), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), an automotive infotainment device, a sailing electronic device (such as a sailing navigation device or a gyro compass), avionics, security devices, vehicular head units, industrial or home robots, automatic teller machines (ATMs), point of sales (POS) devices, or Internet of Things (IoT) devices (such as a bulb, various sensors, electric or gas meter, sprinkler, fire alarm, thermostat, street light, toaster, fitness equipment, hot water tank, heater, or boiler). Other examples of an electronic device include at least one part of a piece of furniture or building/structure, an electronic board, an electronic signature receiving device, a projector, or various measurement devices (such as devices for measuring water, electricity, gas, or electromagnetic waves). Note that, according to various embodiments of this disclosure, an electronic device may be one or a combination of the above-listed devices. According to some embodiments of this disclosure, the electronic device may be a flexible electronic device. The electronic device disclosed here is not limited to the above-listed devices and may include any other electronic devices now known or later developed.

[0016] In the following description, electronic devices are described with reference to the accompanying drawings, according to various embodiments of this disclosure. As used here, the term "user" may denote a human or another device (such as an artificial intelligent electronic device) using the electronic device.

[0017] Definitions for other certain words and phrases may be provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

[0018] None of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claim scope. The scope of patented subject matter is defined only by the claims. Moreover, none of the claims is intended to invoke 35 U.S.C. § 112 (f) unless the exact words "means for" are followed by a participle. Use of any other term, including without limitation "mechanism," "module," "device," "unit," "component," "element," "member," "apparatus," "machine," "system," "processor," or "controller," within a claim is understood by the Applicant to refer to structures known to those skilled in the relevant art and is not intended to invoke 35 U.S.C. § 112 (f).

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] For a more complete understanding of this disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which:

[0020] FIG. **1** illustrates an example network configuration including an electronic device in accordance with this disclosure;

[0021] FIGS. **2**A and **2**B illustrate example errors and corrections associated with a video see-through (VST) extended reality (XR) device in accordance with this disclosure;

[0022] FIG. **3** illustrates a first example functional architecture supporting registration and parallax error correction for VST XR in accordance with this disclosure;

[0023] FIGS. **4**A and **4**B illustrate a second example functional architecture supporting registration and parallax error correction for VST XR in accordance with this disclosure;

[0024] FIG. **5** illustrates an example self-calibration architecture supporting registration and parallax error correction for VST XR in accordance with this disclosure;

[0025] FIG. **6** illustrates example coordinate systems associated with registration and parallax error correction for VST XR in accordance with this disclosure;

[0026] FIGS. **7** through **14** illustrate example corrections of registration and parallax errors for VST XR in accordance with this disclosure; and

[0027] FIG. **15** illustrates an example method for registration and parallax error correction for VST XR in accordance with this disclosure.

DETAILED DESCRIPTION

[0028] FIGS. **1** through **15**, discussed below, and the various embodiments of this disclosure are described with reference to the accompanying drawings. However, it should be appreciated that this disclosure is not limited to these embodiments, and all changes and/or equivalents or replacements thereto also belong to the scope of this disclosure. The same or similar reference denotations may be used to refer to the same or similar elements throughout the specification and the drawings.

[0029] As noted above, extended reality (XR) systems are becoming more and more popular over time, and numerous applications have been and are being developed for XR systems. Some XR systems (such as augmented reality or "AR" systems and mixed reality or "MR" systems) can enhance a user's view of his or her current environment by overlaying digital content (such as information or virtual objects) over the user's view of the current environment. For example, some XR systems can often seamlessly blend virtual objects generated by computer graphics with real-world scenes.

[0030] Optical see-through (OST) XR systems refer to XR systems in which users directly view real-world scenes through head-mounted devices (HMDs). Unfortunately, OST XR systems face many challenges that can limit their adoption. Some of these challenges include limited fields of view, limited usage spaces (such as indoor-only usage), failure to display fully-opaque black objects, and usage of complicated optical pipelines that may require projectors, waveguides, and other optical elements. In contrast to OST XR systems, video see-through (VST) XR systems (also called "passthrough" XR systems) present users with generated video sequences of real-world scenes. VST XR systems can be built using virtual reality (VR) technologies and can have various advantages over OST XR systems. For example, VST XR systems can provide wider fields of view and can provide improved contextual augmented reality.

[0031] VST XR devices typically use see-through cameras to capture images of their surrounding environments. The see-through cameras of a VST XR device are positioned at locations away from a user's eyes, so images of a scene cannot be captured at the locations of the user's eyes. As a result, the images (if presented to the user directly) would provide views of the scene that are different from the views at the user's eyes. Among other things, this can result in mis-registration between objects in the scene as perceived at the user's eyes and the same objects in the scene as captured in the images. Parallax differences can also exist since the viewpoints at the user's eyes are different from the viewpoints at the see-through cameras. Registration and parallax errors in images presented to users may cause the users to feel uncomfortable or even experience motion sickness.

[0032] This disclosure provides various techniques supporting registration and parallax error correction for VST XR. As described in more detail below, a transformation associated with a VST XR device can be identified using a registration error and a parallax error. The registration error and the parallax error can be based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera of the VST XR device and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel of the VST device. An image that captures an object can be obtained using the see-through camera, and the transformation can be applied to the image in order to generate a modified image. The modified image can be rendered for presentation on the display panel. By applying the transformation, the image can be modified such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user. Any suitable number of images may be received and processed in this manner, such as by receiving and processing sequences of images from left and right see-through cameras.

[0033] In this way, these techniques can be used to efficiently and effectively reduce or eliminate registration and

parallax errors in final views of a scene presented to a user of a VST XR device. For example, these techniques allow rendered images presented to a user to appear as if they were captured using see-through cameras located at the positions of the user's eyes, even though the see-through cameras are physically located elsewhere. The overall result is that the final views of the scene can have a higher quality, which can increase user satisfaction and reduce or avoid problems like user discomfort or motion sickness.

[0034] FIG. **1** illustrates an example network configuration **100** including an electronic device in accordance with this disclosure. The embodiment of the network configuration **100** shown in FIG. **1** is for illustration only. Other embodiments of the network configuration **100** could be used without departing from the scope of this disclosure.

[0035] According to embodiments of this disclosure, an electronic device **101** is included in the network configuration **100**. The electronic device **101** can include at least one of a bus **110**, a processor **120**, a memory **130**, an input/output (I/O) interface **150**, a display **160**, a communication interface **170**, and a sensor **180**. In some embodiments, the electronic device **101** may exclude at least one of these components or may add at least one other component. The bus **110** includes a circuit for connecting the components **120-180** with one another and for transferring communications (such as control messages and/or data) between the components.

[0036] The processor **120** includes one or more processing devices, such as one or more microprocessors, microcontrollers, digital signal processors (DSPs), application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs). In some embodiments, the processor **120** includes one or more of a central processing unit (CPU), an application processor (AP), a communication processor (CP), a graphics processor unit (GPU), or a neural processing unit (NPU). The processor **120** is able to perform control on at least one of the other components of the electronic device **101** and/or perform an operation or data processing relating to communication or other functions. As described below, the processor **120** may perform one or more functions related to registration and parallax error correction for VST XR.

[0037] The memory **130** can include a volatile and/or non-volatile memory. For example, the memory **130** can store commands or data related to at least one other component of the electronic device **101**. According to embodiments of this disclosure, the memory **130** can store software and/or a program **140**. The program **140** includes, for example, a kernel **141**, middleware **143**, an application programming interface (API) **145**, and/or an application program (or "application") **147**. At least a portion of the kernel **141**, middleware **143**, or API **145** may be denoted an operating system (OS).

[0038] The kernel **141** can control or manage system resources (such as the bus **110**, processor **120**, or memory **130**) used to perform operations or functions implemented in other programs (such as the middleware **143**, API **145**, or application **147**). The kernel **141** provides an interface that allows the middleware **143**, the API **145**, or the application **147** to access the individual components of the electronic device **101** to control or manage the system resources. The application **147** may include one or more applications that, among other things, perform registration and parallax error correction for VST XR. These functions can be performed by a single application or by multiple applications that each carries out one or more of these functions. The middleware **143** can function as a relay to allow the API **145** or the application **147** to communicate data with the kernel **141**, for instance. A plurality of applications **147** can be provided. The middleware **143** is able to control work requests received from the applications **147**, such as by allocating the priority of using the system resources of the electronic device **101** (like the bus **110**, the processor **120**, or the memory **130**) to at least one of the plurality of applications **147**. The API **145** is an interface allowing the application **147** to control functions provided from the kernel **141** or the middleware **143**. For example, the API **145** includes at least one interface or function (such as a command) for filing control, window control, image processing, or text control.

[0039] The I/O interface **150** serves as an interface that can, for example, transfer commands or data input from a user or other external devices to other component(s) of the electronic device **101**. The I/O interface **150** can also output commands or data received from other component(s) of the electronic device **101** to the user or the other external device.

[0040] The display **160** includes, for example, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a quantum-dot light emitting diode (QLED) display, a microelectromechanical systems (MEMS) display, or an electronic paper display. The display **160** can also be a depth-aware display, such as a multi-focal display. The display **160** is able to display, for example, various contents (such as text, images, videos, icons, or symbols) to the user. The display **160** can include a touchscreen and may receive, for example, a touch, gesture, proximity, or hovering input using an electronic pen or a body portion of the user.

[0041] The communication interface **170**, for example, is able to set up communication between the electronic device **101** and an external electronic device (such as a first electronic device **102**, a second electronic device **104**, or a server **106**). For example, the communication interface **170** can be connected with a network **162** or **164** through wireless or wired communication to communicate with the external electronic device. The communication interface **170** can be a wired or wireless transceiver or any other component for transmitting and receiving signals.

[0042] The wireless communication is able to use at least one of, for example, WiFi, long term evolution (LTE), long term evolution-advanced (LTE-A), 5th generation wireless system (5G), millimeter-wave or 60 GHz wireless communication, Wireless USB, code division multiple access (CDMA), wideband code division multiple access (WCDMA), universal mobile telecommunication system (UMTS), wireless broadband (WiBro), or global system for mobile communication (GSM), as a communication protocol. The wired connection can include, for example, at least one of a universal serial bus (USB), high definition multimedia interface (HDMI), recommended standard 232 (RS-232), or plain old telephone service (POTS). The network **162** or **164** includes at least one communication network, such as a computer network (like a local area network (LAN) or wide area network (WAN)), Internet, or a telephone network.

[0043] The electronic device **101** further includes one or more sensors **180** that can meter a physical quantity or detect an activation state of the electronic device **101** and convert metered or detected information into an electrical signal. For

example, the sensor(s) **180** can include cameras or other imaging sensors, which may be used to capture images of scenes. The sensor(s) **180** can also include one or more buttons for touch input, one or more microphones, a depth sensor, a gesture sensor, a gyroscope or gyro sensor, an air pressure sensor, a magnetic sensor or magnetometer, an acceleration sensor or accelerometer, a grip sensor, a proximity sensor, a color sensor (such as a red green blue (RGB) sensor), a bio-physical sensor, a temperature sensor, a humidity sensor, an illumination sensor, an ultraviolet (UV) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an ultrasound sensor, an iris sensor, or a fingerprint sensor. Moreover, the sensor(s) **180** can include one or more position sensors, such as an inertial measurement unit that can include one or more accelerometers, gyroscopes, and other components. In addition, the sensor(s) **180** can include a control circuit for controlling at least one of the sensors included here. Any of these sensor(s) **180** can be located within the electronic device **101**.

[0044] In some embodiments, the electronic device **101** can be a wearable device or an electronic device-mountable wearable device (such as an HMD). For example, the electronic device **101** may represent an XR wearable device, such as a headset or smart eyeglasses. In other embodiments, the first external electronic device **102** or the second external electronic device **104** can be a wearable device or an electronic device-mountable wearable device (such as an HMD). In those other embodiments, when the electronic device **101** is mounted in the electronic device **102** (such as the HMD), the electronic device **101** can communicate with the electronic device **102** through the communication interface **170**. The electronic device **101** can be directly connected with the electronic device **102** to communicate with the electronic device **102** without involving with a separate network.

[0045] The first and second external electronic devices **102** and **104** and the server **106** each can be a device of the same or a different type from the electronic device **101**. According to certain embodiments of this disclosure, the server **106** includes a group of one or more servers. Also, according to certain embodiments of this disclosure, all or some of the operations executed on the electronic device **101** can be executed on another or multiple other electronic devices (such as the electronic devices **102** and **104** or server **106**). Further, according to certain embodiments of this disclosure, when the electronic device **101** should perform some function or service automatically or at a request, the electronic device **101**, instead of executing the function or service on its own or additionally, can request another device (such as electronic devices **102** and **104** or server **106**) to perform at least some functions associated therewith. The other electronic device (such as electronic devices **102** and **104** or server **106**) is able to execute the requested functions or additional functions and transfer a result of the execution to the electronic device **101**. The electronic device **101** can provide a requested function or service by processing the received result as it is or additionally. To that end, a cloud computing, distributed computing, or client-server computing technique may be used, for example. While FIG. **1** shows that the electronic device **101** includes the communication interface **170** to communicate with the external electronic device **104** or server **106** via the network **162** or **164**, the electronic device **101** may be independently operated without a separate communication function according to some embodiments of this disclosure.

[0046] The server **106** can include the same or similar components as the electronic device **101** (or a suitable subset thereof). The server **106** can support to drive the electronic device **101** by performing at least one of operations (or functions) implemented on the electronic device **101**. For example, the server **106** can include a processing module or processor that may support the processor **120** implemented in the electronic device **101**. As described below, the server **106** may perform one or more functions related to registration and parallax error correction for VST XR.

[0047] Although FIG. **1** illustrates one example of a network configuration **100** including an electronic device **101**, various changes may be made to FIG. **1**. For example, the network configuration **100** could include any number of each component in any suitable arrangement. In general, computing and communication systems come in a wide variety of configurations, and FIG. **1** does not limit the scope of this disclosure to any particular configuration. Also, while FIG. **1** illustrates one operational environment in which various features disclosed in this patent document can be used, these features could be used in any other suitable system.

[0048] FIGS. **2**A and **2**B illustrate example errors and corrections associated with a VST XR device in accordance with this disclosure. For ease of illustration, the VST XR device is described as being implemented using the electronic device **101** in the network configuration **100**. However, the errors and corrections shown in FIGS. **2**A and **2**B may be associated used with any other suitable device(s) and in any other suitable system(s).

[0049] As shown in FIG. **2**A, a scene **202** is being imaged and includes at least one object **204** (a tree in this example). Here, the scene **202** is depicted as being on a plane, which indicates that the scene is being imaged at a constant depth. Note, however, that this is not required, such as when the scene **202** can be imaged at various depths. In this example, the VST XR device includes left and right see-through cameras **206***a*-**206***b*, which can be used to capture images of the scene **202**. The see-through cameras **206***a*-**206***b* may, for example, represent different imaging sensors **180** of the electronic device **101**. Each of the see-through cameras **206***a*-**206***b* can be used to capture see-through images, which represent images that capture a 3D scene from the perspective of that see-through camera **206***a*-**206***b*.

[0050] Images are generated and rendered by the VST XR device for presentation on display panels **208***a*-**208***b* of the VST XR device. The display panels **208***a*-**208***b* may, for example, represent one or more displays **160** of the electronic device **101**. The display panels **208***a*-**208***b* may represent separate displays **160** or different portions of the same display **160**. The rendered images are viewable by the user's eyes **210***a*-**210***b*. As can be seen here, the see-through camera **206***a*-**206***b* are not located at the positions of the user's eyes **210***a*-**210***b*. Instead, the see-through camera **206***a*-**206***b* may be positioned outward or inward of optical axes **212***a*-**212***b* at a distance denoted dx, above or below the optical axes **212***a*-**212***b* at a distance denoted dy, and/or ahead of the user's eyes **210***a*-**210***b* at a distance denoted dz. Note that while the see-through camera **206***a*-**206***b* are displaced from the user's eyes **210***a*-**210***b* by all three distances dx, dy, and dz in FIG. **2**A, the see-through camera

206*a*-206*b* may be displaced from the user's eyes 210*a*-210*b* by one or two of these distances.

[0051] As noted above, the different positions of the see-through cameras 206*a*-206*b* and the user's eyes 210*a*-210*b* can create registration errors and/or parallax errors. For example, a point on the object **204** in FIG. **2** may appear to the user's eye 210*a* as being farther to the left than desired by a distance $error_{left}$, and the same point on the object **204** in FIG. **2** may appear to the user's eye 210*b* as being farther to the right than desired by a distance $error_{right}$. These distances $error_{left}$ and $error_{right}$ are exaggerated for ease of illustration here, but this does illustrate a problem. Registration errors and/or parallax errors can lead to the appearance of two versions 204*a*-204*b* of the object **204**, where those versions 204*a*-204*b* of the object **204** do not overlap. Thus, the user visually sees two different instances of the same object **204**.

[0052] The techniques described below can be used to adjust the images that are presented on the display panels 208*a*-208*b* in order to reduce or eliminate registration errors and/or parallax errors. As a result, these techniques can be used to achieve the results shown in FIG. **2**B, where the rendered images presented on the display panels 208*a*-208*b* are adjusted so that common points of the object **204** are aligned with one another. As a result, the user sees a single version of the object **204** when viewing the display panels 208*a*-208*b*. Techniques for achieving reduction or elimination of registration errors and parallax errors are described below.

[0053] Although FIGS. **2**A and **2**B illustrate examples of errors and corrections associated with a VST XR device, various changes may be made to FIGS. **2**A and **2**B. For example, the scene being viewed and the registration and parallax errors being corrected can vary based on the circumstances.

[0054] FIG. **3** illustrates a first example functional architecture **300** supporting registration and parallax error correction for VST XR in accordance with this disclosure. For ease of explanation, the architecture **300** of FIG. **3** is described as being implemented using the electronic device **101** in the network configuration **100** of FIG. **1**, where the electronic device **101** represents a VST XR device having the various components shown in FIGS. **2**A and **2**B. However, the architecture **300** may be implemented using any other suitable device(s) and in any other suitable system(s).

[0055] As shown in FIG. **3**, the architecture **300** is generally divided into two primary operations, namely a modeling and static transformation creation operation **302** and a dynamic transformation and rendering operation **304**. The modeling and static transformation creation operation **302** generally operates to identify a passthrough transformation that can be used to at least correct for registration and parallax errors. The dynamic transformation and rendering operation **304** generally operates to at least apply the passthrough transformation to obtained see-through images in order to generate rendered images for presentation to a user of the VST XR device, where the rendered images have reduced or minimal registration and parallax errors. In some embodiments, the modeling and static transformation creation operation **302** can provide an efficient and potentially latency-free passthrough transformation for use in rendering the images.

[0056] In this example, the modeling and static transformation creation operation **302** includes a distortion mesh creation function **306**, which generally operates to create a distortion mesh to be used to warp captured images. A distortion mesh represents a mesh of points that defines how images can be transformed or distorted to correct for various issues. In this example, the distortion mesh creation function **306** can be used to create an initial distortion mesh, which can be subsequently modified by the modeling and static transformation creation operation **302**. The distortion mesh creation function **306** can use any suitable technique to generate distortion meshes.

[0057] In some cases, the distortion mesh creation function **306** may create an initial distortion mesh based on one or more extrinsic parameters of at least one see-through camera 206*a*-206*b*, one or more extrinsic parameters of at least one display panel 208*a*-208*b*, and one or more extrinsic parameters of at least one virtual camera. A virtual camera refers to an imaginary camera positioned at the location of one of the user's eyes 210*a*-210*b*, so two virtual cameras can be defined for the user's two eyes 210*a*-210*b*. In some embodiments, there may be two distortion meshes created (one for the see-through camera 206*a*, display panel 208*a*, and left virtual camera and another for the see-through camera 206*b*, display panel 208*b*, and right virtual camera), and each distortion mesh may be used as discussed below. As a particular example, each initial distortion mesh may be based on one or more positions of one or more see-through cameras 206*a*-206*b*, one or more positions of one or more display panels 208*a*-208*b*, and one or more expected positions of one or more of the user's eyes 210*a*-210*b*. Note that these parameters typically do not change as long as the geometric configuration of the VST XR device does not change. Some VST XR devices may, for example, allow modifications to the positions of certain components to support different inter-pupillary distances, which refer to the distance between the user's eyes 210*a*-210*b*. Thus, for instance, as long as the inter-pupillary distance remains constant, there may be no need to recompute the initial distortion mesh(es). However, upon a change in the inter-pupillary distance, the distortion mesh creation function **306** may generate one or more new initial distortion meshes for the current configuration of the VST XR device.

[0058] A registration and parallax error correction function **308** generally operates to transform each initial distortion mesh to correct for various issues within a VST XR device, including registration and parallax errors. For example, the registration and parallax error correction function **308** may determine how captured images would need to be modified in order to correct for possible registration and parallax errors, and the registration and parallax error correction function **308** can modify each initial distortion mesh in order to compensate for the associated registration and parallax errors. As noted above, the user's eyes 210*a*-210*b* may be associated with virtual cameras, and the registration and parallax error correction function **308** may be used to modify each initial distortion mesh based on an estimated/expected registration error and an estimated/expected parallax error. The registration and parallax errors are based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using the see-through camera(s) 206*a*-206*b* of the VST XR device and (ii) one or more perceived positions of the contents of the scene at the virtual camera(s) associated with the viewpoint(s) of the user's eye(s) 210*a*-210*b* when viewing the display panel(s) 208*a*-208*b*. The registration and parallax error

correction function **308** therefore operates to generate a transformation that can correct for the registration and parallax errors.

[0059] In this example, the registration and parallax error correction function **308** includes or has access to a registration error model **310** and a parallax error model **312**. The registration error model **310** generally represents a mathematical model or other model that captures how registration errors may be created between (i) a scene as captured using one or more of the see-through cameras **206***a*-**206***b* and (ii) a scene as viewed by one or more of the user's eyes **210***a*-**210***b* (meaning a scene that would be captured using one or more virtual cameras at the position(s) of the user's eye(s) **210***a*-**210***b*). Similarly, the parallax error model **312** generally represents a mathematical model or other model that captures how parallax errors may be created between (i) the scene as captured using one or more of the see-through cameras **206***a*-**206***b* and (ii) the scene as viewed by one or more of the user's eyes **210***a*-**210***b*. The registration error model **310** and the parallax error model **312** can therefore be used to estimate what the registration and parallax errors would be based on the actual current configuration of the VST XR device and how those errors can be reduced or minimized.

[0060] The registration and parallax error correction function **308** here modifies each initial distortion mesh and generates a corresponding modified distortion mesh. Each modified distortion mesh may optionally be processed using one or more other correction functions **314**, which can further modify the modified distortion mesh in order to correct for other issues. For example, the one or more other correction functions **314** can be used to perform camera undistortion, correct for display lens geometric distortion, and correct for chromatic aberration.

[0061] With respect to camera undistortion, a see-through camera **206***a*-**206***b* or other imaging sensor **180** typically includes at least one lens, and the at least one lens can create radial, tangential, or other type(s) of distortion(s) in captured images. A correction function **314** may make adjustments to each distortion mesh so that the resulting distortion mesh substantially corrects for the radial, tangential, or other type(s) of distortion(s). In some cases, the correction function **314** may include or have access to a camera matrix and lens distortion model, which can be used to identify how each distortion mesh should be adjusted so that the resulting distortion mesh substantially corrects for the camera lens distortion(s). A camera matrix is often defined as a three-by-three matrix that includes two focal lengths in the x and y directions and the principal point of the camera defined using x and y coordinates. A lens distortion model is often defined as a mathematical model that indicates how images can be undistorted, which can be derived based on the specific lens or other optical component(s) being used.

[0062] With respect to display lens geometric distortion correction (GDC) and chromatic aberration correction (CAC), rendered images presented on one or more displays **160** are often viewed by the user through left and right display lenses positioned between the user's eyes **210***a*-**210***b* and the display panels **208***a*-**208***b*. However, the display lenses may create geometric distortions when displayed images are viewed, and the display lenses may create chromatic aberrations when light passes through the display lenses. One or more correction functions **314** generally operate to make adjustments to each distortion mesh so that the resulting distortion mesh compensates for the geometric distortions and the chromatic aberrations. Thus, for instance, the correction function(s) **314** may determine how images should be pre-distorted to compensate for the subsequent geometric distortions and chromatic aberrations created when the images are displayed and viewed through the display lenses. In some cases, the correction function(s) **314** may operate based on a display lens GDC and CAC model, which can mathematically represent the geometric distortions and chromatic aberrations created by the display lenses.

[0063] Each resulting distortion mesh generated using the modeling and static transformation creation operation **302** represents a passthrough transformation **316**. Each passthrough transformation **316** represents or is based on a final distortion mesh and is configured to correct for registration and parallax errors and other issues. Each passthrough transformation **316** can be stored (such as in a memory **130**) and provided to the dynamic transformation and rendering operation **304** when needed.

[0064] In some cases, each passthrough transformation **316** may be defined ahead of time, stored, and applied when see-through image capture starts, such as when each passthrough transformation **316** is generated during an initialization of a VST XR device. In other cases, each passthrough transformation **316** may be defined at the start of see-through image capture, and the same passthrough transformation(s) **316** can be applied throughout see-through image capture. This is because the specific configuration of the VST XR device can generally remain fixed during see-through image capture, so the passthrough transformation(s) **316** can be determined for that specific configuration and applied to the see-through images captured using that specific configuration. Thus, each passthrough transformation **316** may represent a static transformation that can be applied to the see-through images without requiring computational resources or associated latency to keep identifying the passthrough transformation **316** as the see-through images are being captured and processed. Moreover, since each passthrough transformation **316** can already include registration and parallax corrections and optionally other corrections, the final rendered images presented on the display panel(s) **208***a*-**208***b* provide the desired registration and the desired parallax. Note, however, that the passthrough transformations **316** can be easily reidentified as needed or desired using the approaches described here, such as if the positions and/or angles of the see-through cameras **206***a*-**206***b* or other components of the VST XR device are adjusted manually or automatically during use.

[0065] The dynamic transformation and rendering operation **304** includes various functions that capture, process, and render images for presentation to the user of the VST XR device. An image and depth data capture function **318** generally operates to capture images and (optionally) depth data associated with the images for processing. For example, the image and depth data capture function **318** may obtain images captured using one or more see-through cameras **206***a*-**206***b* or other imaging sensors **180**. The image and depth data capture function **318** may also optionally obtain depth maps or other depth data. In some embodiments, depth maps or other depth data may be obtained using a depth sensor or other sensor(s) **180** of the VST XR device or by performing depth reconstruction in which depth values in a scene are derived based on stereo images of the scene (where

disparities in locations of common points in the stereo images are used to estimate depths). As a particular example, each depth map may be generated by obtaining an initial depth map and increasing the resolution of the initial depth map using depth super-resolution and depth verification operations, which is often referred to as "densification."

[0066] A dynamic passthrough transformation function **320** generally operates to transform the captured images based on (among other things) the passthrough transformation(s) **316** and the optional depth data. For example, the dynamic passthrough transformation function **320** may apply one or more passthrough transformations **316** to the captured images in order to adjust the captured images to correct for registration and parallax errors. The dynamic passthrough transformation function **320** may also apply one or more additional transformations to the images, such as a head pose change compensation transformation. For example, there is typically a delay between capture of images and display of corresponding rendered images, and it is possible for the user to move his or her head during that intervening time period. The dynamic passthrough transformation function **320** can use information (such as from an IMU, tracking camera, or other data) to predict how the user's head pose is expected to change between capture of the images and display of the corresponding rendered images. The dynamic passthrough transformation function **320** can also apply a suitable transformation to each captured image, such as by rotating and/or translating each captured image, so that the transformed images are suitable for presentation at the user's predicted head pose. The dynamic passthrough transformation function **320** may further perform any additional color corrections or other modifications to generate transformed images.

[0067] The transformed images generated by the dynamic passthrough transformation function **320** are provided to a final view rendering function **322**, which generally operates to process the transformed images and perform any additional refinements or modifications needed or desired. The resulting images can represent the final views of the scene, and the final view rendering function **322** can present rendered images of the final views to the user. For example, a 3D-to-2D warping can be used to warp the final views of the scene into 2D images. The final view rendering function **322** can render the images into a form suitable for transmission to at least one display **160** and can initiate display of the rendered images, such as by providing the rendered images to one or more displays **160**.

[0068] Although FIG. **3** illustrates a first example of a functional architecture **300** supporting registration and parallax error correction for VST XR, various changes may be made to FIG. **3**. For example, various components or functions in FIG. **3** may be combined, further subdivided, replicated, omitted, or rearranged and additional components or functions may be added according to particular needs.

[0069] FIGS. **4**A and **4**B illustrate a second example functional architecture **400** supporting registration and parallax error correction for VST XR in accordance with this disclosure. For ease of explanation, the architecture **400** of FIGS. **4**A and **4**B is described as being implemented using the electronic device **101** in the network configuration **100** of FIG. **1**, where the electronic device **101** represents a VST XR device having the various components shown in FIGS. **2**A and **2**B. However, the architecture **400** may be implemented using any other suitable device(s) and in any other suitable system(s).

[0070] As shown in FIG. **4**A, the architecture **400** includes or has access to various types of headset layout information **402**, which represents information associated with the current configuration of the VST XR device. For example, the headset layout information **402** may include see-through camera configuration data **404**, virtual camera configuration data **406**, and display panel configuration data **408**. Among other things, the see-through camera configuration data **404** defines the location of each see-through camera **206***a*-**206***b* of the VST XR device, the virtual camera configuration data **406** defines the location of each virtual camera, and the display panel configuration data **408** defines the location of each display panel **208***a*-**208***b* of the VST XR device. The location of each virtual camera can be based on the actual or expected position of the associated eye **210***a*-**210***b* of the user, which may be identified based on the design or current configuration of the VST XR device, the current interpupillary distance of the VST XR device, or input from one or more sensors **180**.

[0071] The architecture **400** also includes or has access to various types of camera calibration information **410**, which represents information associated with each see-through camera **206***a*-**206***b*. For example, the camera calibration information **410** may include a camera matrix **412** for each see-through camera **206***a*-**206***b*, a lens model **414** for each see-through camera **206***a*-**206***b*, one or more extrinsic parameters **416** for each see-through camera **206***a*-**206***b*, and stereo camera pair calibration data **418**. As noted above, the camera matrix **412** for each see-through camera **206***a*-**206***b* may be defined as a three-by-three matrix that includes two focal lengths of the see-through camera in the x and y directions and the principal point of the see-through camera defined using x and y coordinates. The lens model **414** for each see-through camera **206***a*-**206***b* may represent a lens distortion model, which as noted above is often defined as a mathematical model that indicates how images can be undistorted (such as based on the specific lens or other optical component(s) being used in the see-through camera). The one or more extrinsic parameters **416** for each see-through camera **206***a*-**206***b* relate to the location and orientation of the see-through camera, such as by identifying the translation and rotation of the see-through camera using six degrees of freedom (like three translation values along three orthogonal axes and three rotation values about the three orthogonal axes). The stereo camera pair calibration data **418** includes data related to how a stereo pair of see-through cameras **206***a*-**206***b* are related to one another, such as geometrically. As described above, depth data can be derived using stereo images of a scene, where disparities in locations of common points in the stereo images are used to estimate depths. The stereo camera pair calibration data **418** provides information that allows depth data to be derived using stereo images from the stereo pair of see-through cameras **206***a*-**206***b*.

[0072] A distortion mesh creation operation **420** uses the headset layout information **402** to create initial distortion meshes. For example, the distortion mesh creation operation **420** includes a mesh generation function **422**, which generally operates to use the various types of headset layout information **402** to (i) determine an initial distortion mesh for transforming images captured using the see-through

camera **206***a* for display on the display panel **208***a* and viewing by the user's left eye **210***a* and (ii) determine an initial distortion mesh for transforming images captured using the see-through camera **206***b* for display on the display panel **208***b* and viewing by the user's right eye **210***b*. Each initial distortion mesh here can be based on (among other things) the positions of the associated see-through camera **206***a*-**206***b* and the associated display panel **208***a*-**208***b* or virtual camera.

[0073] The distortion mesh creation operation **420** also includes an optional mesh tile and resolution configuration function **424**. In some embodiments, the VST XR device supports foveation rendering, which refers to a process in which the area of an image receiving a user's focus has higher resolution or otherwise higher quality, while areas of the image not receiving the user's focus have lower resolution or otherwise lower quality. When foveation rendering is supported, the mesh tile and resolution configuration function **424** can identify where the user is focusing (such as based on one or more eye tracking cameras or other sensors **180**), and the mesh tile and resolution configuration function **424** can identify different tiles or other portions of each initial distortion mesh and a resolution for each tile or other portion of each initial distortion mesh. For instance, the mesh tile and resolution configuration function **424** may identify one tile or other portion of each initial distortion mesh where the user is focused and assign a higher resolution to those tiles, and the mesh tile and resolution configuration function **424** may identify one or more tiles or other portions of each initial distortion mesh where the user is not focused and assign a lower resolution to those tiles.

[0074] An undistortion operation **426** uses the camera calibration information **410** to determine how captured images may need to be undistorted. As noted above, a see-through camera **206***a*-**206***b* or other imaging sensor **180** typically includes at least one lens that can create radial, tangential, or other type(s) of distortion(s) in captured images. The undistortion operation **426** includes an undistortion and transformation function **428**, which generally operates to process at least some of the camera calibration information **410** and adjust each initial distortion mesh from the distortion mesh creation operation **420** in order to pre-compensate for the radial, tangential, or other type(s) of distortion(s). This allows the resulting distortion meshes to substantially correct for the distortion(s) created by the camera lenses. The undistortion operation **426** also includes a stereo camera rectification function **430**, which generally operates to process at least some of the camera calibration information **410** and determine how images captured using the stereo pair of see-through cameras **206***a*-**206***b* should be modified in order to rectify the images. This allows the stereo camera rectification function **430** to adjust each initial distortion mesh from the distortion mesh creation operation **420** in order to ensure that images captured using the stereo pair of see-through cameras **206***a*-**206***b* are suitably rectified and therefore able to be used in estimating depths within a scene.

[0075] A registration model creation operation **432** is used to generate and apply a registration error model, such as the registration error model **310**. The registration model creation operation **432** includes a see-through camera layout and configuration acquisition function **434**, which generally operates to obtain information about the layout and configuration of the see-through cameras **206***a*-**206***b* of the VST XT device. For example, the acquisition function **434** may obtain at least some of the headset layout information **402** and the camera calibration information **410**. This information can be used to identify the locations of the see-through cameras **206***a*-**206***b*, such as relative to each other and/or to the associated virtual cameras at the locations of the user's eyes **210***a*-**210***b*. A registration error identification function **436** determines registration errors based on the layout and configuration of the see-through cameras **206***a*-**206***b*. For instance, the registration error identification function **436** can identify the registration error between each see-through camera **206***a*-**206***b* and its associated virtual camera, which can be based on the geometry/current configuration of the VST XR device. A registration error correction function **438** determines how to reduce or minimize the identified registration error, such as by identifying an error correction (like a translation, a rotation, or both) that can reduce or minimize the registration error.

[0076] Similarly, a parallax model creation operation **440** is used to generate and apply a parallax error model, such as the parallax error model **312**. The parallax model creation operation **440** includes a see-through camera layout and configuration acquisition function **442**, which generally operates to obtain information about the layout and configuration of the see-through cameras **206***a*-**206***b* of the VST XT device. For example, the acquisition function **442** may obtain at least some of the headset layout information **402** and the camera calibration information **410**. This information can be used to identify the locations of the see-through cameras **206***a*-**206***b*, such as relative to each other and/or to the associated virtual cameras at the locations of the user's eyes **210***a*-**210***b*. Note that the acquisition function **442** may or may not represent the same function as the acquisition function **434**. A parallax error identification function **444** determines parallax errors based on the layout and configuration of the see-through cameras **206***a*-**206***b*. For instance, the parallax error identification function **444** can identify the parallax error between each see-through camera **206***a*-**206***b* and its associated virtual camera, which can be based on the geometry/current configuration of the VST XR device. A parallax error correction function **446** determines how to reduce or minimize the identified parallax error, such as by identifying an error correction (like a translation, a rotation, or both) that can reduce or minimize the parallax error.

[0077] A registration and parallax error correction operation **448** generally operates to modify the distortion meshes from the undistortion operation **426** based on the registration error model and the parallax error model. For example, the registration and parallax error correction operation **448** includes a camera extrinsic parameter acquisition function **450**, a display panel extrinsic parameter acquisition function **452**, and a virtual camera extrinsic parameter acquisition function **454**. The acquisition functions **450**-**454** obtain one or more extrinsic parameters for each of the see-through cameras **206***a*-**206***b*, each of the display panels **208***a*-**208***b*, and each of the virtual cameras. A distortion mesh transformation function **456** uses the acquired information to modify and apply and the error corrections determined using the registration and parallax error models. For instance, the distortion mesh transformation function **456** can use the extrinsic parameters of the see-through cameras **206***a*-**206***b*, display panels **208***a*-**208***b*, and virtual cameras to determine how the error corrections determined using the registration and parallax error models should be transformed so that the

desired error corrections are achieved given the current configuration of the VST XR device.

[0078] As shown in FIG. **4**B, the distortion meshes generated by the distortion mesh transformation function **456** are provided to a display correction operation **458**, which generally operates to provide display-related corrections. For example, the display correction operation **458** includes a GDC function **460**, which provides geometric distortion correction. The display correction operation **458** also includes a CAC function **462**, which provides chromatic aberration correction. As noted above, these corrections may be needed when display lenses positioned between the user's eyes **210***a*-**210***b* and the display panels **208***a*-**208***b* create geometric distortions and chromatic aberrations. In some cases, the GDC function **460** and the CAC function **462** may each use a display lens GDC and CAC model, which can mathematically model the geometric distortions and chromatic aberrations created by the display lenses. The distortion meshes as modified by the display correction operation **458** may represent final transformed distortion meshes to be applied to captured images.

[0079] A data capture operation **464** generally operates to obtain images and related data to be processed. In this example, the data capture operation **464** includes a see-through camera image capture function **466**, which generally operates to obtain images of a scene captured using one or more see-through cameras **206***a*-**206***b* or other imaging sensors **180** of the VST XR device. For instance, the image capture function **466** may be used to obtain see-through images at a specified frame rate. A depth sensor data capture function **468** may be used to obtain depth maps or other depth-related data, such as from one or more depth sensors **180** of the VST XR device. A tracking camera image capture function **470** may be used to obtain tracking images from a tracking camera. A position sensor data capture function **472** may be used to obtain position-related data, such as from an IMU or other position sensor **180** of the VST XR device. The tracking images and the position-related data can relate to the head pose of the user of the VST XR device, which typically varies over time.

[0080] A data mapping operation **474** generally operates to transform the captured data from the data capture operation **464** based on (among other things) the final transformed distortion meshes from the display correction operation **458**. For example, an image-to-distortion mesh mapping function **476** can map each image captured using the see-through cameras **206***a*-**206***b* onto the corresponding final transformed distortion mesh. If foveation rendering is supported, an image-to-foveation mesh mapping function **478** can map each image captured using the see-through cameras **206***a*-**206***b* onto the corresponding final transformed distortion mesh, where the final transformed distortion mesh supports tiles or other areas with different resolutions. A depth data mapping function **480** may map the depth maps or other depth data onto the corresponding final transformed distortion meshes. This effectively provides transformation of the captured images (and optionally other data like the depth data) based on the final transformed distortion meshes, which among other things helps to correct for registration and parallax errors.

[0081] A reprojection operation **482** generally operates to reproject the transformed images generated by the data mapping operation **474** to account for user head pose changes. As described above, there is typically a delay between capture of images and display of corresponding rendered images, and it is possible for the user to move his or her head during that intervening time period. The reprojection operation **482** includes a head pose data acquisition function **484** and a user focus data acquisition function **486**. The head pose data acquisition function **484** generally operates to obtain information related to the user's head pose, such as the user's current head pose when each captured image is obtained and the user's predicted head pose when the corresponding rendered image is predicted to be displayed. The user's predicted head pose can be estimated in any suitable manner, such as by using a head pose model to estimate what the user's future head pose might be given the user's current head pose and recent movements. The user focus data acquisition function **486** generally operates to obtain information related to the user's current area of focus, which is also called the user's region of interest. A head pose change compensation function **488** generally operates to reproject each of the transformed images generated by the data mapping operation **474** (if necessary) from the head pose of the user when the corresponding original image was captured and the estimated head pose of the user when a rendered version of the transformed image will be displayed. In some cases, the head pose change compensation function **488** can apply a translation and/or a rotation to each transformed image. This can result in the generation of final views of the scene being imaged.

[0082] A final view rendering operation **490** can process the final views and perform any additional image processing functions as needed or desired. For example, a final view enhancement function **492** may perform denoising or other post-processing functions to help increase the quality of the final views. A final view display function **494** can present rendered versions of the final views to the user. For instance, the final view display function **494** can render images into a form suitable for transmission to at least one display **160** and can initiate display of the rendered images, such as by providing the rendered images to one or more displays **160**.

[0083] Although FIGS. **4**A and **4**B illustrate a second example of a functional architecture **400** supporting registration and parallax error correction for VST XR, various changes may be made to FIGS. **4**A and **4**B. For example, various components or functions in FIGS. **4**A and **4**B may be combined, further subdivided, replicated, omitted, or rearranged and additional components or functions may be added according to particular needs. As a particular example, the registration and parallax models are described as being used to support transformations performed by the data mapping operation **474** to correct registration and parallax errors. However, the registration and parallax models may be used to support transformations performed elsewhere, such as transformations performed by the final view rendering operation **490**. In other words, transformations used for registration and parallax error correction can be performed in various locations within the architecture **400**. As another particular example, the architecture **400** is designed based on the assumption that the transformations for correcting registration and parallax errors are generally static. However, in other cases, the registration and parallax models can be generated using or otherwise based on depth information within a scene (which can vary dynamically),

and registration and parallax correction transformations can be applied based on dynamic registration and parallax models.

[0084] FIG. **5** illustrates an example self-calibration architecture **500** supporting registration and parallax error correction for VST XR in accordance with this disclosure. In some embodiments, the self-calibration architecture **500** can be used with or included in the architecture **300** or **400**, such as to self-register each see-through camera **206***a*-**206***b* with its associated virtual camera. As described below, this can be accomplished by integrating a registration error self-calibration and a parallax error self-calibration. This effectively allows the architecture **300** or **400** to modify rendering parameters associated with the virtual cameras based on feedback representing or associated with rendered images. In some cases, the self-calibration architecture **500** may not need to be used continuously and may be used periodically or intermittently, such as in response to a head pose change by the user of the VST device.

[0085] As shown in FIG. **5**, the architecture **500** receives and processes various input data **502**, such as see-through camera images **504**, depth maps **506**, and head position data **508**. For example, the see-through camera images **504** may be obtained using the see-through camera image capture function **466**, the depth maps **506** may be obtained using the depth sensor data capture function **468**, and the head position data **508** may be obtained using the tracking camera image capture function **470** and the position sensor data capture function **472**.

[0086] The obtained information is provided to a transformation and rendering operation **510**, which generally operates to transform the captured images and render the resulting transformed images. In this example, the transformation and rendering operation **510** includes the data mapping operation **474**, the reprojection operation **482**, and the final view rendering operation **490** discussed above. The transformation and rendering operation **510** also includes a head pose prediction function **512**, which generally operates to process the head position data **508** and make predictions about the head pose of the user. These predictions can be used by the reprojection operation **482** as discussed above to compensate for head pose changes by the user. The final view rendering operation **490** here can use rendering parameters **514** associated with the virtual cameras when rendering images for presentation to the user.

[0087] In this example, a self-calibration operation **516** can be performed using the rendered images **518** produced by the transformation and rendering operation **510**, and the self-calibration operation **516** can be used to update the parameters **514** associated with the virtual cameras for use by the final view rendering operation **490**. For instance, the self-calibration operation **516** includes or has access to the registration error model **310** and the parallax error model **312**. A registration error identification function **520** can determine a registration error between a current see-through camera image **504** and the rendered image **518** associated with that image **504** using the registration error model **310**. In some cases, the depth map **506** associated with the current see-through camera image **504** can be used when determining the registration error. This represents part of a registration error self-calibration. Similarly, a parallax registration error identification function **522** can determine a parallax error between the current see-through camera image **504** and the rendered image **518** associated with that image **504** using the parallax error model **312**. In some cases, the depth map **506** associated with the current see-through camera image **504** can be used when determining the parallax error. This represents part of a parallax error self-calibration. As can be seen here, the rendered image **518** acts as feedback.

[0088] The identified registration and parallax errors are provided to a virtual camera self-calibration function **524**, which generally operates to update the parameters **514** associated with the virtual cameras based the identified registration and parallax errors between the current see-through camera image **504** and the rendered image **518** associated with that image **504**. For example, the virtual camera self-calibration function **524** can determine how to adjust the parameters **514** associated with the virtual cameras in order to reduce the registration and parallax errors. The virtual camera self-calibration function **524** can adjust the parameters **514** associated with the virtual cameras in any suitable manner, such as by adjusting the parameters **514** to apply a translation and/or a rotation to the parameters **514** in order to reduce the registration and parallax errors.

[0089] A parameter update function **526** provides the updated parameters **514** to the final view rendering operation **490** for use in re-rendering the image **504**. This results in the generation of another rendered image **518**, which can again be used as feedback. As can be seen here, this approach supports both a registration error self-calibration and a parallax error self-calibration. The self-calibration process can be repeated until the registration error and/or the parallax error is less that a corresponding threshold.

[0090] Although FIG. **5** illustrates one example of a self-calibration architecture **500** supporting registration and parallax error correction for VST XR, various changes may be made to FIG. **5**. For example, various components or functions in FIG. **5** may be combined, further subdivided, replicated, omitted, or rearranged and additional components or functions may be added according to particular needs.

[0091] FIG. **6** illustrates example coordinate systems associated with registration and parallax error correction for VST XR in accordance with this disclosure. These coordinate systems can be associated with a VST XR device, and these coordinate systems may be used as discussed below during registration and parallax error correction.

[0092] As shown in FIG. **6**, a global coordinate system **600** represents a 3D coordinate system in the real world and is defined using three axes X, Y, and Z that meet at an origin O. A camera coordinate system **602** is associated with a see-through camera **206***a*-**206***b* and is defined using three axes X_c, Y_c, and Z_c that meet at an origin Oc. An image coordinate system **604** is associated with a see-through camera image **504** and is defined using two axes x_i and y_i that meet at an origin O_i (which may be located at the center of the image **504**). A pixel coordinate system **606** is associated with pixels within a see-through camera image **504** and is defined using two axes x_c and y_c that meet at an origin O_p (which may be located at the top left corner or other corner of the image **504**). A point of an object **204** within a 3D scene being captured is denoted as $D_{cam}(X_{cam}, Y_{cam}, Z_{cam})$ within the global coordinate system **600**. That point appears within the image **504** at a point denoted $D_c(X_c, Y_c, Z_c)$ in the camera coordinate system **602**, a point denoted $d_i(x_i, y_i)$ in the image coordinate system **604**, and a point denoted $d_c(x_c, y_c)$ in the pixel coordinate system **606**.

[0093] The relationships between the 3D point D_{cam} in the scene to the point D_c in the camera coordinate system **602** and an image pixel at point d_c in the pixel coordinate system **606** can be defined as follows. Assume pixels of the see-through camera image **504** are generated based on the intrinsic and extrinsic parameters of the associated see-through camera **206***a*-**206***b*. Assume K_c is the camera matrix of the see-through camera **206***a*-**206***b*, where the camera matrix is expressed as follows.

$$K_c = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

[0094] Here, (f_x, f_y) is the focal length of the see-through camera **206***a*-**206***b*, and (c_x, c_y) is the principal point of the see-through camera **206***a*-**206***b*. From this, a relationship or transformation **608** can be defined between the 3D point D_{cam} in the global coordinate system **600** and the camera point D_c in the camera coordinate system **602**, such as by defining the following relationship.

$$D_c(X_c, Y_c, Z_c, W_c) = \begin{pmatrix} R_c & t_c \\ 0 & 1 \end{pmatrix} D_{cam}(X_{cam}, Y_{cam}, Z_{cam}, W_{cam})$$

Here, R_c represents a rotation matrix of the see-through camera **206***a*-**206***b*, and t_c represents a translation vector of the see-through camera **206***a*-**206***b*. A relationship or transformation **610** can also be defined between the camera point D_c in the camera coordinate system **602** and the corresponding point d_c in the pixel coordinate system **606**, such as by defining the following relationship.

$$\lambda_c d_c(x_c, y_c, w_c) = K_c D_c(X_c, Y_c, Z_c)$$

Here, λ_c represents a scale factor. Rewriting these equations, the following relationship can be defined between the 3D point D_{cam} in the global coordinate system **600** and the corresponding point d_c in the pixel coordinate system **606**.

$$\lambda_c d_c(x_c, y_c, w_c) = K_c[R_c|t_c] D_{cam}(X_{cam}, Y_{cam}, Z_{cam})$$

Here, P_c is a projection matrix and may be defined as follows.

$$P_c = K_c[R_c|t_c]$$

These types of relationships can be used to support corrections of various registration and parallax errors for VST XR as described below.

[0095] Although FIG. **6** illustrates examples of coordinate systems **600**-**606** associated with registration and parallax error correction for VST XR, various changes may be made to FIG. **6**. For example, each coordinate system **600**-**606** can define their axes and origin in any suitable manner.

[0096] FIGS. **7** through **14** illustrate example corrections of registration and parallax errors for VST XR in accordance with this disclosure. These corrections may, for example, be identified and implemented using the architecture **300** of FIG. **3** or the architecture **400** of FIGS. **4**A and **4**B described above. Note that the see-through camera **206***a*, display panel **208***a*, and user's eye **210***a* are mentioned in the following discussion and shown in FIGS. **7** through **14**. The same or similar errors and corrections may involve the see-through camera **206***b*, display panel **208***b*, and user's eye **210***b*.

[0097] As shown in FIG. **7**, the see-through camera **206***a* is offset laterally from the user's eye **210***a*, meaning dx is non-zero while dy and dz are substantially equal to zero. This can lead to the creation of a registration/parallax error in the x direction. For example, a point D_{cam} of the object **204** appears as a point d_e, which can be based on the image pixel at point d_c as determined by the following.

$$\lambda_e d_e(x_e, y_e, w_e) = H_{ce} d_c(x_c, y_c, w_c)$$

Here, H_{ce} represents a homographic transformation matrix between the see-through camera **206***a* and the virtual camera at the user's eye **210***a*, which in some cases may be expressed as follows.

$$H_{ce} = K_e\left(R_{ce} + \frac{t_{ce} n_c^T}{d_{cp}}\right) K_c^{-1}$$

Here, K_e represents the camera matrix of the virtual camera. Since a transformation is desired from the see-through camera **206***a* to the virtual camera, the following can be obtained.

$$K_e = K_c$$

Here, R_{ce} represents a rotation matrix between the see-through camera **206***a* and the virtual camera, t_{ce} represents a translation vector between the see-through camera **206***a* and the virtual camera, n_c represents a normal vector of an image plane **702**, and d_{cp} represents a distance between the see-through camera **206***a* and the image plane **702**. Based on this, the following can be obtained.

$$\begin{pmatrix} x_e \\ y_e \\ w_e \end{pmatrix} = \frac{1}{\lambda_e} H_{ce} \begin{pmatrix} x_c \\ y_c \\ w_c \end{pmatrix}$$

[0098] Since both the see-through camera **206***a* and the virtual camera have the same orientation in FIG. **7**, $R_{ce}=I$, where I represents an identity matrix. Thus, the virtual camera only has a translation d_x relative to the see-through camera **206***a*. As a result, the following can be obtained.

$$t_{ce} = \begin{pmatrix} d_x \\ 0 \\ 0 \end{pmatrix}$$

The image plane **702** is perpendicular to the see-through camera **206***a*, so the following can be obtained.

$$n_c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Based on this, the homographic transformation matrix H_{ce} can be rewritten as follows.

$$H_{ce} = K_c \begin{pmatrix} 1 & 0 & \frac{d_x}{d_{cp}} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} K_c^{-1}$$

The relationship between the virtual camera and the 3D scene can be expressed as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix}$$

Here, P_e represents a projection matrix, which could be expressed as follows.

$$P_e = K_e[R_e|t_e]$$

where:

$$\begin{cases} K_e = K_c \\ R_e = R_c \end{cases}$$

$$t_e = \begin{pmatrix} t_{xc} + d_x \\ t_{yc} \\ t_{zc} \end{pmatrix}$$

The location of the 3D point as captured at the virtual camera could be expressed as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_e \\ y_e \\ 1 \\ we \end{pmatrix}$$

Based on this, the projection matrix P_e could be defined as follows.

$$P_e = K_c[R_c|t_e]$$

[0099] As shown in FIG. **7**, if the virtual camera image point $d_e(x_e, y_e)$ is matched to the same location of the corresponding see-through camera image point $d_c(x_c, y_c)$, the virtual camera would render the same image as the see-through camera image. However, this creates an error $error_x$ between the actual position of the 3D point D_{cam} and the perceived position of that point D_{eye} created by presentation of the virtual camera image point d_e. In this condition, the location of the 3D point as captured at the virtual camera could be rewritten as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_c \\ y_c \\ 1 \\ wc \end{pmatrix}$$

Here, the points D_{cam} and D_{eye} do not overlap, which can be easily noticeable by a user. Since there is only a shift in the x direction between the see-through camera **206***a* and the virtual camera, the error can be defined as follows.

$$error_x - X_{eye} - X_{cam}$$

[0100] To register the points D_{cam} and D_{eye} generated by the see-through camera **206***a* and the virtual camera and make them overlap as shown in FIG. **8**, the location of the eye view image point d_c can be adjusted by a specified error correction $p_{correct}$. In this example, the point d_c and the point d_e will not overlap one another, but the points D_{cam} and D_{eye} will substantially overlap and thereby resolve the error. In some cases, the error correction $p_{correct}$ may be determined as follows.

$$p_{correct} = x_c - x_e$$

From this, the location of the eye view image point d_e can be determined as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ W_{eya} \end{pmatrix}$$

[0101] As shown in FIG. **9**, the see-through camera **206***a* is offset vertically from the user's eye **210***a*, meaning dy is non-zero while dx and dz are substantially equal to zero. This can lead to the creation of a registration/parallax error in the y direction. For example, a point D_{cam} of the object **204** appears as a point d_e, which can be based on the image pixel at point d_c as determined by the following.

$$\lambda_e d_e(x_e, t_e, w_e) = H_{ce} d_c(x_c, y_c, w_c)$$

Here, H_{ce} represents a homographic transformation matrix between the see-through camera **206***a* and the virtual camera at the user's eye **210***a*, which in some cases may be expressed as follows.

$$H_{ce} = K_e \left(R_{ce} + \frac{t_{ce} n_c^T}{d_{cp}} \right) K_c^{-1}$$

[0102] Here, K_e represents the camera matrix of the virtual camera. Since a transformation is desired from the see-through camera **206***a* to the virtual camera, the following can be obtained.

$$K_e = K_c$$

Here, R_{ce} represents a rotation matrix between the see-through camera **206***a* and the virtual camera, t_{ce} represents a translation vector between the see-through camera **206***a* and the virtual camera, n_c represents a normal vector of the image plane **702**, and d_{cp} represents a distance between the see-through camera **206***a* and the image plane **702**. Based on this, the following can be obtained.

$$\begin{pmatrix} x_e \\ y_e \\ w_e \end{pmatrix} = \frac{1}{\lambda_e} H_{ce} \begin{pmatrix} x_c \\ y_c \\ w_c \end{pmatrix}$$

[0103] Since both the see-through camera **206***a* and the virtual camera have the same orientation in FIG. **9**, R_{ce}=I, where I represents an identity matrix. Thus, the virtual camera only has a translation dy relative to the see-through camera **206***a*. As a result, the following can be obtained.

$$t_{ce} = \begin{pmatrix} 0 \\ d_y \\ 0 \end{pmatrix}$$

The image plane **702** is perpendicular to the see-through camera **206***a*, so the following can be obtained.

$$n_c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Based on this, the homographic transformation matrix H_{ce} can be rewritten as follows.

$$H_{ce} = K_c \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{d_y}{d_{cp}} \\ 0 & 0 & 1 \end{pmatrix} K_c^{-1}$$

The relationship between the virtual camera and the 3D scene can be expressed as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix}$$

Here, P_e represents a projection matrix, which could be expressed as follows.

$$P_e = K_e[R_e \mid t_e]$$

where:

$$\begin{cases} K_e = K_c \\ R_e = R_c \end{cases}$$

$$t_e = \begin{pmatrix} t_{xc} \\ t_{yc} + d_y \\ t_{zc} \end{pmatrix}$$

The location of the 3D point as captured at the virtual camera could be expressed as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_e \\ y_e \\ 1 \\ we \end{pmatrix}$$

Based on this, the projection matrix P_e could be defined as follows.

$$P_e = K_c[R_c \mid t_e]$$

[0104] As shown in FIG. **9**, if the virtual camera image point $d_e(x_e, y_e)$ is matched to the same location of the corresponding see-through camera image point $d_c(x_c, y_c)$, the virtual camera would render the same image as the see-through camera image. However, this creates an error $error_y$ between the actual position of the 3D point D_{cam} and the perceived position of that point D_{eye} created by presentation of the virtual camera image point d_e. In this condition, the location of the 3D point as captured at the virtual camera could be rewritten as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_c \\ y_c \\ 1 \\ wc \end{pmatrix}$$

Here, the points D_{cam} and D_{eye} do not overlap, which can be easily noticeable by a user. Since there is only a shift in the y direction between the see-through camera **206***a* and the virtual camera, the error can be defined as follows.

$$error_y = Y_{eye} - Y_{cam}$$

To register the points D_{cam} and D_{eye} generated by the see-through camera **206***a* and the virtual camera and make them overlap as shown in FIG. **10**, the location of the eye view image point d_c can be adjusted by a specified error correction $p_{correct}$. In this example, the point d_c and the point d_c will not overlap one another, but the points D_{cam} and D_{eye} will substantially overlap and thereby resolve the error. In some cases, the error correction $p_{correct}$ may be determined as follows.

$$p_{correct} = y_c - y_e$$

From this, the location of the eye view image point d_c can be determined as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ W_{eya} \end{pmatrix}$$

[0105] As shown in FIGS. **11** and **13**, the see-through camera **206***a* is offset axially from the user's eye **210***a*,

meaning dz is non-zero while dx and dy are substantially equal to zero. This can lead to the creation of registration/parallax errors in both the x and y directions. For example, in both the x and y directions, a point D_{cam} of the object **204** appears as a point d_e, which can be based on the image pixel at point d_c as determined by the following.

$$\lambda_e d_e(x_e, y_e, w_e) = H_{ce} d_c(x_c, y_c, w_c)$$

Here, H_{ce} represents a homographic transformation matrix between the see-through camera **206***a* and the virtual camera at the user's eye **210***a*, which in some cases may be expressed as follows.

$$H_{ce} = K_e \left(R_{ce} + \frac{t_{ce} n_c^T}{d_{cp}} \right) K_c^{-1}$$

Here, K_e represents the camera matrix of the virtual camera. Since a transformation is desired from the see-through camera **206***a* to the virtual camera, the following can be obtained.

$$K_e = K_c$$

Here, R_{ce} represents a rotation matrix between the see-through camera **206***a* and the virtual camera, t_{ce} represents a translation vector between the see-through camera **206***a* and the virtual camera, n_c represents a normal vector of the image plane **702**, and d_{cp} represents a distance between the see-through camera **206***a* and the image plane **702**. Based on this, the following can be obtained.

$$\begin{pmatrix} x_e \\ y_e \\ w_e \end{pmatrix} = \frac{1}{\lambda_e} H_{ce} \begin{pmatrix} x_c \\ y_c \\ w_c \end{pmatrix}$$

[0106] Since both the see-through camera **206***a* and the virtual camera have the same orientation in FIGS. **11** and **13**, R_{ce}=I, where I represents an identity matrix. Thus, the virtual camera only has a translation dz relative to the see-through camera **206***a*. As a result, the following can be obtained.

$$t_{ce} = \begin{pmatrix} 0 \\ 0 \\ d_z \end{pmatrix}$$

The image plane **702** is perpendicular to the see-through camera **206***a*, so the following can be obtained.

$$n_c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Based on this, the homographic transformation matrix H_{ce} can be rewritten as follows.

$$H_{ce} = K_c \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{d_z}{d_{cp}} \end{pmatrix} K_c^{-1}$$

The relationship between the virtual camera and the 3D scene can be expressed as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix}$$

Here, P_e represents a projection matrix, which could be expressed as follows.

$$P_e = K_e [R_e | t_e]$$

where:

$$\begin{cases} K_e = K_c \\ R_e = R_c \end{cases}$$

$$t_e = \begin{pmatrix} t_{xc} \\ t_{yc} \\ t_{zc} + d_z \end{pmatrix}$$

The location of the 3D point as captured at the virtual camera could be expressed as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_c \\ y_e \\ 1 \\ we \end{pmatrix}$$

Based on this, the projection matrix P_e could be defined as follows.

$$P_e = K_c [R_c | t_e]$$

[0107] As shown in FIG. **12**, if the virtual camera image point $d_e(x_e, y_e)$ is matched to the same location of the corresponding see-through camera image point $d_c(x_c, y_c)$, the virtual camera would render the same image as the see-through camera image. However, this creates an error $error_{zx}$ between the actual position of the 3D point D_{cam} and the perceived position of that point D_{eye} created by presentation of the virtual camera image point d_e. In this condition, the location of the 3D point as captured at the virtual camera could be rewritten as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_c \\ y_c \\ 1 \\ wc \end{pmatrix}$$

Here, the points D_{cam} and D_{eye} do not overlap, which can be easily noticeable by a user. Since there is only a shift in the y direction between the see-through camera **206***a* and the virtual camera, the error can be defined as follows.

$$error_{zx} = X_{eye} - X_{cam}$$

To register the points D_{cam} and D_{eye} generated by the see-through camera **206***a* and the virtual camera and make them overlap as shown in FIG. **12**, the location of the eye view image point d_e can be adjusted by a specified error correction $p_{x\_correct}$. In this example, the point d_c and the point d_e will not overlap one another, but the points D_{cam} and D_{eye} will substantially overlap and thereby resolve the error in the x direction. In some cases, the error correction $p_{x\_correct}$ may be determined as follows.

$$P_{x\_correct} = x_c - x_e$$

From this, the location of the eye view image point d_e can be determined as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ W_{eya} \end{pmatrix}$$

[0108] Similarly, as shown in FIG. **14**, if the virtual camera image point $d_e(x_e, y_e)$ is matched to the same location of the corresponding see-through camera image point $d_c(x_c, y_c)$, the virtual camera would render the same image as the see-through camera image. However, this creates an error $error_{zy}$ between the actual position of the 3D point D_{cam} and the perceived position of that point D_{eye} created by presentation of the virtual camera image point d_e. In this condition, the location of the 3D point as captured at the virtual camera could be rewritten as follows.

$$\begin{pmatrix} X_e \\ Y_e \\ Z_e \\ W_e \end{pmatrix} = \lambda_e P_e^{-1} \begin{pmatrix} x_c \\ y_c \\ 1 \\ wc \end{pmatrix}$$

Here, the points D_{cam} and D_{eye} do not overlap, which can be easily noticeable by a user. Since there is only a shift in the y direction between the see-through camera **206***a* and the virtual camera, the error can be defined as follows.

$$error_{zy} = Y_{eye} - Y_{cam}$$

[0109] To register the points D_{cam} and D_{eye} generated by the see-through camera **206***a* and the virtual camera and make them overlap as shown in FIG. **12**, the location of the eye view image point d_e can be adjusted by a specified error correction $p_{y\_correct}$. In this example, the point d_c and the point d_e will not overlap one another, but the points D_{cam} and D_{eye} will substantially overlap and thereby resolve the error in the y direction. In some cases, the error correction $p_{y\_correct}$ may be determined as follows.

$$P_{y\_correct} = y_c - y_e$$

From this, the location of the eye view image point d_e can be determined as follows.

$$\lambda_e \begin{pmatrix} x_e \\ y_e \\ 1 \\ w_e \end{pmatrix} = P_e \begin{pmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ W_{eya} \end{pmatrix}$$

[0110] Although FIGS. **7** through **14** illustrate examples of corrections of registration and parallax errors for VST XR, various changes may be made to FIGS. **7** through **14**. For example, FIGS. **7** through **14** have illustrated errors caused by translation of a see-through camera **206***a* relative to a user's eye in one direction (x, y, or z translation). However, the see-through camera **206***a* may be translated relative to a user's eye in multiple directions, and/or the see-through camera **206***a* may be rotated relative to the axis of a user's eye when looking straight ahead. The equations and mathematical derivations shown above can be modified to account for these other configurations of the see-through camera **206***a*.

[0111] FIG. **15** illustrates an example method **1500** for registration and parallax error correction for VST XR in accordance with this disclosure. For ease of explanation, the method **1500** of FIG. **15** is described as being performed using the electronic device **101** in the network configuration **100** of FIG. **1**, where the electronic device **101** can implement the architecture **300** of FIG. **3** or the architecture **400** of FIGS. **4**A and **4**B. However, the method **1500** may be performed using any other suitable device(s) and architecture(s) and in any other suitable system(s).

[0112] As shown in FIG. **15**, a transformation associated with a VST XR device is identified at step **1502**. This could include, for example, the processor **120** of the electronic device **101** generating a transformation using a registration error and a parallax error. The registration error and the parallax error can be based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera **206***a*-**206***b* of the electronic device **101** and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel **208***a*-**208***b* of the electronic device **101**. In some cases, a registration error model **310** can be used to identify the registration error, and a parallax error model **312** (separate from the registration error model **310**) can be used to identify the parallax error. In some embodiments, the transformation is static and is based on the configuration of the electronic device **101**.

[0113] An image captured using the see-through camera of the VST XR device is obtained at step **1504**. This could include, for example, the processor **120** of the electronic device **101** obtaining a see-through camera image **504** captured using the see-through camera **206***a*-**206**. The image is transformed by applying the identified transformation to the image at step **1506**. This could include, for example, the processor **120** of the electronic device **101** applying the identified transformation to correct for registration and parallax errors. This results in the generation of a modified image.

[0114] The modified image is rendered at step **1508**. This could include, for example, the processor **120** of the electronic device **101** rendering the modified image for display.

A self-registration of the see-through camera and the virtual camera may optionally be performed at step **1510**. This could include, for example, the processor **120** of the electronic device **101** using the rendered image as feedback (such as within the self-calibration architecture **500**) and possibly adjusting parameters of the virtual camera used to render the modified image. If necessary, step **1508** can be repeated one or more times based on the feedback. Presentation of the rendered image can be initiated at step **1512**. This could include, for example, the processor **120** of the electronic device **101** presenting the final version of the rendered image on the display panel **208***a*-**208***b*.

[0115] Although FIG. **15** illustrates one example of a method **1500** for registration and parallax error correction for VST XR, various changes may be made to FIG. **15**. For example, while shown as a series of steps, various steps in FIG. **15** may overlap, occur in parallel, occur in a different order, or occur any number of times (including zero times). Also, various steps in FIG. **15** may be repeated in order to process any suitable number of images from any suitable number of see-through cameras, such as to process a sequence of images from left and right see-through cameras **206***a*-**206***b*.

[0116] It should be noted that the functions described above can be implemented in an electronic device **101**, **102**, **104**, server **106**, or other device(s) in any suitable manner. For example, in some embodiments, at least some of the functions can be implemented or supported using one or more software applications or other software instructions that are executed by the processor **120** of the electronic device **101**, **102**, **104**, server **106**, or other device(s). In other embodiments, at least some of the functions can be implemented or supported using dedicated hardware components. In general, the functions described above can be performed using any suitable hardware or any suitable combination of hardware and software/firmware instructions. Also, the functions described above can be performed by a single device or by multiple devices.

[0117] Although this disclosure has been described with example embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that this disclosure encompass such changes and modifications as fall within the scope of the appended claims.

What is claimed is:

1. A method comprising:

identifying a transformation associated with a video see-through (VST) extended reality (XR) device using a registration error and a parallax error, the registration error and the parallax error based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera of the VST XR device and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel of the VST device;

obtaining an image that captures an object using the see-through camera;

applying the transformation to the image in order to generate a modified image; and

rendering the modified image for presentation on the display panel;

wherein applying the transformation modifies the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

2. The method of claim **1**, further comprising:

self-registering the see-through camera and the virtual camera by integrating (i) a self-calibration of the virtual camera based on the registration error and (ii) a self-calibration of the virtual camera based on the parallax error.

3. The method of claim **2**, wherein:

self-registering the see-through camera and the virtual camera updates one or more parameters of the virtual camera; and

the one or more updated parameters of the virtual camera are used to re-render the modified image.

4. The method of claim **2**, wherein self-registering the see-through camera and the virtual camera repeats until at least one of an updated registration error or an updated parallax error is less than a threshold.

5. The method of claim **1**, wherein identifying the transformation comprises:

using a registration model to identify the registration error; and

using a parallax model to identify the parallax error, the parallax model separate from the registration model.

6. The method of claim **1**, wherein the transformation comprises a static transformation.

7. The method of claim **1**, wherein the transformation is based on one or more extrinsic parameters of the see-through camera, one or more extrinsic parameters of the display panel, and one or more extrinsic parameters of the virtual camera.

8. A video see-through (VST) extended reality (XR) device comprising:

a see-through camera;

a display panel; and

at least one processing device configured to:

identify a transformation using a registration error and a parallax error, the registration error and the parallax error based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using the see-through camera and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing the display panel;

obtain an image that captures an object using the see-through camera;

apply the transformation to the image in order to generate a modified image; and

render the modified image for presentation on the display panel;

wherein the at least one processing device is configured to apply the transformation in order to modify the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

9. The VST XR device of claim **8**, wherein the at least one processing device is further configured to self-register the see-through camera and the virtual camera by integrating (i) a self-calibration of the virtual camera based on the registration error and (ii) a self-calibration of the virtual camera based on the parallax error.

10. The VST XR device of claim 9, wherein:
the at least one processing device is configured to update one or more parameters of the virtual camera as a result of self-registering the see-through camera and the virtual camera; and
the at least one processing device is configured to re-render the modified image using the one or more updated parameters.

11. The VST XR device of claim 9, wherein the at least one processing device is configured to repeatedly self-register the see-through camera and the virtual camera until at least one of an updated registration error or an updated parallax error is less than a threshold.

12. The VST XR device of claim 8, wherein, to identify the transformation, the at least one processing device is configured to:
use a registration model to identify the registration error; and
use a parallax model to identify the parallax error, the parallax model separate from the registration model.

13. The VST XR device of claim 8, wherein the transformation comprises a static transformation.

14. The VST XR device of claim 8, wherein the transformation is based on one or more extrinsic parameters of the see-through camera, one or more extrinsic parameters of the display panel, and one or more extrinsic parameters of the virtual camera.

15. A non-transitory machine readable medium containing instructions that when executed cause at least one processor of a video see-through (VST) extended reality (XR) device to:
identify a transformation using a registration error and a parallax error, the registration error and the parallax error based on one or more differences between (i) one or more actual positions of contents of a scene as imaged using a see-through camera of the VST XR device and (ii) one or more perceived positions of the contents of the scene at a virtual camera associated with a viewpoint of a user when viewing a display panel of the VST device;
obtain an image that captures an object using the see-through camera;
apply the transformation to the image in order to generate a modified image; and
render the modified image for presentation on the display panel;
wherein the instructions when executed cause the at least one processor to apply the transformation in order to modify the image such that a perceived position of the object substantially matches an actual position of the object when the display panel is viewed by the user.

16. The non-transitory machine readable medium of claim 15, further containing instructions that when executed cause the at least one processor to:
self-register the see-through camera and the virtual camera by integrating (i) a self-calibration of the virtual camera based on the registration error and (ii) a self-calibration of the virtual camera based on the parallax error.

17. The non-transitory machine readable medium of claim 16, wherein:
the instructions when executed cause the at least one processor to update one or more parameters of the virtual camera as a result of self-registering the see-through camera and the virtual camera; and
the instructions when executed cause the at least one processor to re-render the modified image using the one or more updated parameters.

18. The non-transitory machine readable medium of claim 16, wherein the instructions when executed cause the at least one processor to repeatedly self-register the see-through camera and the virtual camera until at least one of an updated registration error or an updated parallax error is less than a threshold.

19. The non-transitory machine readable medium of claim 15, wherein the instructions that when executed cause the at least one processor to identify the transformation comprise:
instructions that when executed cause the at least one processor to:
use a registration model to identify the registration error; and
use a parallax model to identify the parallax error, the parallax model separate from the registration model.

20. The non-transitory machine readable medium of claim 15, wherein the transformation is based on one or more extrinsic parameters of the see-through camera, one or more extrinsic parameters of the display panel, and one or more extrinsic parameters of the virtual camera.

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